Ordinary Differential Equations

Shuo Guo

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Contents

One could read *Differential Equations and Their Applications* to have a better understanding of this notes:
https://link.springer.com/book/10.1007/978-1-4612-4360-1

Chapter 1

First Order Differential Equations

For the first order differential equations, we would consider to solve

$$\frac{\mathrm{d}y}{\mathrm{d}t} = f(t, y)$$

Consider a simpler case:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = g(t)$$

One with elementary calculus knowledge can solve it:

$$y(t) = \int g(t)dt + c$$

However, for general first order equation, it's very hard to solve.
We would first consider some simpler cases.

1 First order linear differential equations

Consider f is consisted by linear equations.

Definition 1.1.1 (First-oder linear differential equation).

The general first-order linear differential equation is

$$\frac{\mathrm{d}y}{\mathrm{d}t} + a(t)y = b(t) \tag{1}$$

Needless to say, $a(t)$, $b(t)$ are assumed to be continuous functions of time. To slove general first order linear equation(1), we could consider a simpler case:

Definition 1.1.2 (Homogeneous first order linear equation).
The equation

$$\frac{\mathrm{d}y}{\mathrm{d}t} + a(t)y = 0 \tag{2}$$

is called homogeneous first-order linear differential equation

1.1 Homogeneous first order linear differential equation

Theorem 1.1.1. *For equation in the form:*

$$\frac{\mathrm{d}y}{\mathrm{d}t} + a(t)y = 0,$$

the solution is

$$y(t) = c\, e^{-\int a(t)\, dt}$$

Moreover, if we have $y(t_0) = y_0$, the solution is

$$y(t) = y(t_0)\exp(\int_{t_0}^{t} a(s)ds)$$

Proof.

$$\frac{\mathrm{d}y}{\mathrm{d}t} + a(t)y = 0.$$
$$\frac{\mathrm{d}y}{\mathrm{d}t} = -a(t)y.$$
$$\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}t} = -a(t).$$

Integrating both sides with respect to t gives

$$\int \frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}t}\, dt = -\int a(t)\, dt.$$

By the chain rule, $\frac{1}{y}\frac{dy}{dt} = \frac{d}{dt}\ln|y|$, hence

$$\ln|y(t)| = -\int a(t)\, dt + C,$$

where C is an arbitrary constant of integration. Exponentiating both sides yields

$$|y(t)| = e^{C} e^{-\int a(t)\, dt}.$$

Since $e^C > 0$, we may absorb the sign into a new constant $c = \pm e^C$, obtaining the general solution

$$\boxed{y(t) = c\, e^{-\int a(t)\, dt}} \tag{3}$$

where $c \in \mathbb{R}$ is an arbitrary constant. Note that the zero solution corresponds to $c = 0$.

Now we add the condition:

$$y(t_0) = y_0 \tag{4}$$

Then we integral (3) between t_0 and t.
Recalling $\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}t} = -a(t)$, we could write:

$$\int_{t_0}^{t} \frac{\mathrm{d}}{\mathrm{d}s}\ln|y(s)|ds = \int_{t_0}^{t} -a(s)ds$$

So
$$\ln|y(t)| - \ln|y(t_0)| = \ln|\frac{y(t)}{y(t_0)}| = -\int_{t_0}^{t} a(s)ds$$
Thus,
$$|\frac{y(t)}{y(t_0}| = \exp(-\int_{t_0}^{t} a(s)ds)$$
The problem now becomes +1 or -1. By taking $t = t_0$, we can find it's +1.
So
$$y(t) = y(t_0)\exp(\int_{t_0}^{t} a(s)ds) \tag{5}$$
□

1.2 General first order linear differential equation

Theorem 1.1.2. *For differential equations with the form:*

$$\frac{\mathrm{d}y}{\mathrm{d}t} + a(t)y = b(t) \tag{6}$$

the solution is

$$y = \frac{1}{\mu(t)}(\int \mu(t)b(t)dt + c), \quad \textit{with } \mu(t) = \exp(\int a(t)dt).$$

Moreover, if we have $y_0 = y(t_0)$, then we have:

$$y = \frac{1}{\mu(t)}\Big(\mu(t_0)y_0 + \int_{t_0}^{t} \mu(s)b(s)dt\Big)$$

Proof. We want to write it in a form:

$$\frac{\mathrm{d}'something'}{\mathrm{d}t} = c(t)$$

and then we can solve it. Multiply both sides by a continuous function $\mu(t)$ [1]
So

$$\mu(t)\frac{\mathrm{d}y}{\mathrm{d}t} + a(t)\mu(t)y = \mu(t)b(t)$$

We consider another formula: $\frac{\mathrm{d}}{\mathrm{d}t}\mu(t)y = \frac{\mathrm{d}y}{\mathrm{d}t}\mu(t) + y\frac{\mathrm{d}\mu(t)}{\mathrm{d}t}$.
So we try to choose proper μ s.t. the left of the equation can be written as $\frac{\mathrm{d}'something'}{\mathrm{d}t}$.
We choose

$$\mu(t) = \exp(\int a(t)dt) \tag{7}$$

And

$$\frac{\mathrm{d}}{\mathrm{d}t}\mu(t)y = \mu(t)b(t)$$

$$\mu(t)y = \int \mu(t)b(t)dt + c$$

$$y = \frac{1}{\mu(t)}(\int \mu(t)b(t)dt + c), \quad \text{with } \mu(t) = \exp(\int a(t)dt) \tag{8}$$

If $y(t_0) = y_0$: So:

$$\frac{\mathrm{d}}{\mathrm{d}t}\mu(t)y = \mu(t)b(t) \quad \Rightarrow \quad \mu(t)y - \mu(t_0)y_0 = \int_{t_0}^{t} \mu(s)b(s)ds$$

Thus,

$$y = \frac{1}{\mu(t)}\Big(\mu(t_0)y_0 + \int_{t_0}^{t} \mu(s)b(s)dt\Big) \tag{9}$$

□

[1] Here any continuous μ could satisfies the condition.

2 Separable equations

Recall how we deal with linear differential equation:

$$\frac{1}{y(t)}\frac{\mathrm{d}y}{\mathrm{d}t} = -a(t)$$

$$\Downarrow$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\ln|g(t)| = -a(t)$$

So for a more general ODE with the form:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{g(t)}{f(y)} \tag{10}$$

We could write it as

$$\frac{\mathrm{d}}{\mathrm{d}t}F(y(t)) = g(t), \quad \text{with } F(y) = \int f(y)dy$$

So the answer is:

$$F(y(t)) = \int g(t)dt + c \tag{11}$$

We could also add an assumption $y(t_0) = y_0$
So

$$F(y(t)) - F(y(t_0)) = \int_{t_0}^{t} g(s)ds$$

$$\int_{y_0}^{y} f(r)dr = \int_{t_0}^{t} g(s)ds$$

3 Population models

Let $p(t)$ denotes the population of a given species at time t, $r(t, p)$ denotes the difference between its birth rate and death rate.
For the simplest, we think that r will not change with t and p, which means it's a constant. The simplest model:

$$\frac{\mathrm{d}p(t)}{\mathrm{d}t} = ap(t), \quad a \text{ is a constant}$$

Also, assume $p(t_0) = p_0$
We can easily find the solution

$$p(t) = p_0 \exp(a(t - t_0)) \tag{12}$$

This result works very well if the population is small. However, if the population grows larger and larger, it falls. So we can let

$$\frac{\mathrm{d}r}{\mathrm{d}t} = ar - br^2 \tag{13}$$

It's also a sparable problem, though the integer part is a bit complex (but absolutly solvable!)
The answer is:

$$p(t) = \frac{ap_0 e^{a(t-t_0)}}{a - bp_0 + bp_0 e^{a(t-t_0)}} = \frac{ap_0}{bp_0 + (a - bp_0)e^{-a(t-t_0)}}.$$

4 Exact equations and the range we can solve

Consider all differential equation we've solved, we can put them in this form:

$$\frac{\mathrm{d}}{\mathrm{d}t}\phi(t,y) = 0 \tag{14}$$

Example 1.4.1 (ϕ for first order linear differential equations)**.**
Thinking about all sparable differential equation (it would also contains all first order linear differential equations) we've solved before:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = g(t)h(y(t))$$

We can all write this equation in this form:

$$\phi(t,y) = \int^{y} \frac{1}{h(s)}ds - \int^{t} g(r)dr$$

More generally, we can solve all differential equations which can be written in this form:

$$\frac{\mathrm{d}}{\mathrm{d}t}\phi(t,y(t)) = 0 \tag{15}$$

Remark 4.1. Actually, 'we can solve' doesn't mean that we can get explict solution. It just becoma a equation without differential term. But with the aid of computer, we can solve its numer with any desired accuracy.

Consider:

$$\frac{\mathrm{d}}{\mathrm{d}t}\phi(t,y) = \frac{\partial\phi}{\partial t} + \frac{\partial\phi}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t} \tag{16}$$

One natural question is that for all $M(t,y), N(t,y)$, does there exist ϕ s.t.

$$M = \frac{\partial\phi}{\partial t} \quad N = \frac{\partial\phi}{\partial y} \tag{17}$$

The answer is **NO**.

Theorem 1.4.2. *Let $M(t,y)$ and $N(t,y)$ be continuous and have continuous partial derivatives with respect to t and y in the rectangle R consisting of those points (t,y) with $a < t < b$ and $c < y < d$. There exists a function $\phi(t,y)$ such that*

$$M(t,y) = \frac{\partial\phi}{\partial t} \quad \textit{and} \quad N(t,y) = \frac{\partial\phi}{\partial y}$$

if, and only if,

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t}$$

in R.

The entire proof will be in 4.3.

Definition 1.4.1 (Exact equations)**.**
The differential equation

$$M(t,y)+N(t,y)\frac{dy}{dt}=0$$

is said to be exact *if*

$$\frac{\partial M}{\partial y}=\frac{\partial N}{\partial t}.$$

For exact equations, we can write:

$$\phi(t,y)=\int M(t,y)+\int\Big(N(t,y)-\int\frac{\partial M(t,y)}{\partial y}dt\Big)dy \tag{18}$$

Remark 4.3. It's not necessary that $\frac{\partial M}{\partial y}=\frac{\partial N}{\partial t}$ in a rectangle R as theorem 1.4.2 states. It can be any region without 'hole' inside.

Remark 4.4. For differential equation $dy/dt-f(t,y)$, it can be written in the form $M(t,y)+N(t,y)(dy/dt)=0$ by setting $M(t,y)=-f(t,y)$ and $N(t,y)=1$.

Now we've solved Exact equations. Suppose

$$M(t,y)+N(t,y)\frac{\mathrm{d}y}{\mathrm{d}t}=0 \tag{19}$$

is not exact. Could we solve it?
A natural idea is to multiply both terms with a function $\mu(t,y)$:

$$M(t,y)\mu(t,y)+\mu(t,y)N(t,y)\frac{\mathrm{d}y}{\mathrm{d}t}=0 \tag{20}$$

And if μ is well-chosen, it could become an exact equation, and we can have

$$\frac{\partial}{\partial y}\Big(M(t,y)\mu(t,y)\Big)=\frac{\partial}{\partial t}\Big(N(t,y)\mu(t,y)\Big) \tag{21}$$

Definition 1.4.2 (Integrating Factor)**.**
Any function $\mu(t,y)$ satisfying (21) is called integrating factor for the equation (19)

Then we try to find μ.

$$\frac{\partial}{\partial y}\Big(M(t,y)\mu(t,y)\Big)=\frac{\partial}{\partial t}\Big(N(t,y)\mu(t,y)\Big)$$

$$\Downarrow$$

$$M\frac{\mathrm{d}}{\mathrm{d}y}\mu+\mu\frac{\mathrm{d}}{\mathrm{d}y}M=N\frac{\mathrm{d}\mu}{\mathrm{d}t}+\mu\frac{\mathrm{d}}{\mathrm{d}t}N$$

$$\Downarrow$$

$$-N\frac{\mathrm{d}\mu}{\mathrm{d}t} + M\frac{\mathrm{d}\mu}{\mathrm{d}y} = (\frac{\mathrm{d}N}{\mathrm{d}t} - \frac{\mathrm{d}M}{\mathrm{d}y})\mu$$

Solving a general $\mu(t, y)$ is almost as difficult as solving the ODE equation.
So if μ could only depend on t or y, we could be able to solve it. **It's just an assmuption. We cannot say that there must exists $\mu = \mu(t)$ or $\mu = \mu(y)$. Just for some special cases, we calculate some μ below and then we find if we are lucky enough to obtain $\mu = \mu(t)$ or $\mu = \mu(y)$.**

4.1 $\mu = \mu(t)$

If μ depends only on t, then $\partial\mu/\partial y = 0$. Therefore,

$$-N\frac{d\mu}{dt} = \left(\frac{\partial N}{\partial t} - \frac{\partial M}{\partial y}\right)\mu,$$

and hence

$$\frac{1}{\mu}\frac{d\mu}{dt} = \frac{\dfrac{\partial M}{\partial y} - \dfrac{\partial N}{\partial t}}{N}. \tag{22}$$

Since the left-hand side depends only on t, the expression on the right-hand side must also be a function of t alone. If

$$\frac{\dfrac{\partial M}{\partial y} - \dfrac{\partial N}{\partial t}}{N} = f(t),$$

then integrating (22) gives

$$\boxed{\mu(t) = C\exp\left(\int f(t)\,dt\right)}, \qquad C \neq 0. \tag{23}$$

4.2 $\mu = \mu(y)$

If μ depends only on y, then $\partial\mu/\partial t = 0$. Therefore,

$$M\frac{d\mu}{dy} = \left(\frac{\partial N}{\partial t} - \frac{\partial M}{\partial y}\right)\mu,$$

and hence

$$\frac{1}{\mu}\frac{d\mu}{dy} = \frac{\dfrac{\partial N}{\partial t} - \dfrac{\partial M}{\partial y}}{M}. \tag{24}$$

Since the left-hand side depends only on y, the expression on the right-hand side must also be a function of y alone. If

$$\frac{\dfrac{\partial N}{\partial t} - \dfrac{\partial M}{\partial y}}{M} = g(y),$$

then integrating (24) gives

$$\mu(y) = C \exp\left(\int g(y)\, dy\right), \qquad C \neq 0. \tag{25}$$

4.3 Proof of theorem1.4.2

Proof. Fix $t_0 \in (a, b)$. First, suppose that ϕ satisfies

$$\frac{\partial \phi}{\partial t} = M(t, y).$$

Integrating with respect to t, while regarding y as a constant, gives

$$\phi(t, y) = \int_{t_0}^{t} M(s, y)\, ds + h(y), \text{ where } h(y) \text{ is an arbitrary function depending only on } y. \tag{26}$$

Differentiating (26) with respect to y, we obtain

$$\frac{\partial \phi}{\partial y} = \int_{t_0}^{t} \frac{\partial M}{\partial y}(s, y)\, ds + \frac{dh}{dy}(y).$$

We also require $\partial\phi/\partial y = N(t, y)$. Therefore,

$$\frac{dh}{dy}(y) = N(t, y) - \int_{t_0}^{t} \frac{\partial M}{\partial y}(s, y)\, ds. \tag{27}$$

The left-hand side of (27) depends only on y, so the right-hand side must also be independent of t. Differentiating the right-hand side with respect to t, we find

$$\frac{\partial}{\partial t}\left(N(t, y) - \int_{t_0}^{t} \frac{\partial M}{\partial y}(s, y)\, ds\right) = \frac{\partial N}{\partial t}(t, y) - \frac{\partial M}{\partial y}(t, y) = 0$$

Thus, if such a function ϕ exists, then necessarily

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t}.$$

Conversely, if this equality holds throughout R, then the right-hand side of (27) has zero derivative with respect to t, and hence depends only on y. We can therefore integrate it with respect to y to obtain a function $h(y)$. Substituting this h into (26) gives a function satisfying

$$\frac{\partial \phi}{\partial t} = M(t, y) \quad \text{and} \quad \frac{\partial \phi}{\partial y} = N(t, y).$$

Consequently, such a function ϕ exists if and only if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t}.$$

□

5 The existence-uniqueness theorem: Picard iteration

Many first order differential equations (actually, most) we cannot get the explict solution. However, can we prove that there exists a solution? If so, could we prove that it's the only sulution, or it's the only solution under some conditions?

Consider

$$\frac{\mathrm{d}y}{\mathrm{d}t} = f(t, y), \quad y(t_0) = y_0 \tag{28}$$

So

$$y(t) = y_0 + \int_{t_0}^{t} f(s, y(s))ds \quad \text{as } y(t) \text{ is continuous and } y(t_0) = y_0 \tag{29}$$

Define an operator[2]

$$\mathcal{L}[y](t) = y_0 + \int_{t_0}^{t} f(s, y(s))ds$$

$$\mathcal{L} : C([0, T]) \to C([0, T])$$

For equation

$$y(t) = \mathcal{L}[y](t)$$

we call it *fixed point* of $\mathcal{L}$.
So the problem: Finding the solution of ODE → Finding the fixed point of $\mathcal{L}$. We'll use a sequence of $\{y_n\}$ to approximate.

In the following subsubsections, we will learn how to construct a sequence of $\{y_n\}$ in 5.1. We'll talk about whether the sequence will converge in 5.2 (But we don't care where it converges). We'll prove that the convergence result is the solution of (29) in 5.3. Last, we'll prove the uniqueness of the solution in 5.4

5.1 Construction of Picard iterates

To solve

$$y(t) = y_0 + \int_{t_0}^{t} f(s, y(s))ds \quad \text{as } y(t) \text{ is continuous and } y(t_0) = y_0 \tag{30}$$

we create a sequance of $\{y(t)\}$ to approximate the answer. We start with the constant function

$$y_0(t) = y_0.$$

Substituting this initial guess into the right-hand side of the integral equation gives the first Picard iterate:

$$y_1(t) = \mathcal{L}[y_0](t) = y_0 + \int_{t_0}^{t} f(s, y_0(s))\, ds = y_0 + \int_{t_0}^{t} f(s, y_0)\, ds.$$

[2]Actually I still don't get fully understand why we do so complex here. It may have some connection with the space? Anyway, see it, accept it, and see what's going on later.

If $y_1(t) = y_0(t)$, then y_0 is already a fixed point of $\mathcal{L}$, so it is a solution of the initial value problem. If not, we substitute y_1 into $\mathcal{L}$ again and obtain

$$y_2(t) = \mathcal{L}[y_1](t) = y_0 + \int_{t_0}^{t} f(s, y_1(s))\,ds.$$

We continue this process. At each step, if

$$y_{n+1}(t) = y_n(t),$$

then y_n is a fixed point of $\mathcal{L}$ and hence a solution. Otherwise, we use y_{n+1} to construct the next iterate. In general, the Picard iterates are defined by

$$\boxed{y_0(t) = y_0, \qquad y_{n+1}(t) = y_0 + \int_{t_0}^{t} f(s, y_n(s))\,ds, \quad n \geq 0.} \tag{31}$$

Notice that every iterate satisfies $y_n(t_0) = y_0$.
Then we'll show the convergence of Picard iterates.

5.2 Convergence of Picard iterates

Lemma 5.1. *Choose any two positive numbers a and b, and let R be the rectangle:*

$$t_0 \leq t \leq t_0 + a, \quad |y - y_0| \leq b.$$

Compute

$$M = \max_{(t,y)\in R} |f(t, y)|, \quad \textit{and set } \alpha = \min\left(a, \frac{b}{M}\right).$$

Then,

$$|y_n(t) - y_0| \leq M(t - t_0)$$

for $t_0 \leq t \leq t_0 + \alpha$.

Proof. For $n = 0$,

$$|y_0(t) - y_0| = 0 \leq M(t - t_0).$$

Suppose the result is true for $n = j$. Since $t - t_0 \leq \alpha$,

$$|y_j(t) - y_0| \leq M(t - t_0) \leq M\alpha \leq b.$$

Thus, $(t, y_j(t)) \in R$, so $|f(t, y_j(t))| \leq M$. Therefore,

$$\begin{aligned}
|y_{j+1}(t) - y_0| &= \left|\int_{t_0}^{t} f(s, y_j(s))\,ds\right| \\
&\leq \int_{t_0}^{t} |f(s, y_j(s))|\,ds \\
&\leq M(t - t_0).
\end{aligned}$$

Hence the result follows for every $n \geq 0$ by induction. $\square$

We now show the convergence of $\{y_n\}$. First, we write y_n in this form:

$$y_n(t) = y_0(t) + [y_1(t) - y_0(t)] + \cdots + [y_n(t) - y_{n-1}(t)]$$

So the sequence $y_n(t)$ converges iff the infinite series converges:

$$[y_1(t) - y_0(t)] + [y_2(t) - y_1(t)] + \cdots + [y_n(t) - y_{n-1}(t)] + \cdots$$

Then, we try to prove a stronger result:

$$\sum_{n=1}^{\infty} |y_n(t) - y_{n-1}(t)| < \infty \Rightarrow \sum_{n=1}^{\infty} \big[y_n(t) - y_{n-1}(t)\big] \text{ converges.} \tag{32}$$

This is not obvious (at least not obvious for me). The result used in (32) will be proved in 5.5.

$$L = \max_{(t,y)\in R} \left| \frac{\partial f}{\partial y}(t, y) \right|.$$

By the mean value theorem, for some $\xi_n(s)$ between $y_n(s)$ and $y_{n-1}(s)$,

$$\begin{aligned} |f(s, y_n(s)) - f(s, y_{n-1}(s))| &= \left| \frac{\partial f}{\partial y}(s, \xi_n(s)) \right| |y_n(s) - y_{n-1}(s)| \\ &\le L|y_n(s) - y_{n-1}(s)|. \end{aligned}$$

where

$$L = \max_{(t,y)\in R} \left| \frac{\partial f}{\partial y}(t, y) \right|.$$

First,

$$\begin{aligned} |y_1(t) - y_0(t)| &= \left| \int_{t_0}^{t} f(s, y_0)\, ds \right| \\ &\le M(t - t_0). \end{aligned}$$

Then

$$\begin{aligned} |y_2(t) - y_1(t)| &= \left| \int_{t_0}^{t} \big[f(s, y_1(s)) - f(s, y_0(s))\big]\, ds \right| \\ &\le L \int_{t_0}^{t} |y_1(s) - y_0(s)|\, ds \\ &\le ML \int_{t_0}^{t} (s - t_0)\, ds \\ &= \frac{ML(t - t_0)^2}{2!}, \end{aligned}$$

and

$$\begin{aligned} |y_3(t) - y_2(t)| &\leq L \int_{t_0}^{t} |y_2(s) - y_1(s)|\, ds \\ &\leq \frac{ML^2(t-t_0)^3}{3!}. \end{aligned}$$

Continuing in the same way,

$$\boxed{|y_n(t) - y_{n-1}(t)| \leq \frac{ML^{n-1}(t-t_0)^n}{n!}, \qquad n \geq 1.} \tag{33}$$

If $L = 0$, then $y_2 = y_1$. For $L > 0$ and $t_0 \leq t \leq t_0 + \alpha$,

$$\begin{aligned} \sum_{n=1}^{\infty} |y_n(t) - y_{n-1}(t)| &\leq \sum_{n=1}^{\infty} \frac{ML^{n-1}(t-t_0)^n}{n!} \\ &= \frac{M}{L} \sum_{n=1}^{\infty} \frac{[L(t-t_0)]^n}{n!} \\ &= \frac{M}{L} \left(e^{L(t-t_0)} - 1\right) \\ &\leq \boxed{\frac{M}{L}\left(e^{\alpha L} - 1\right)} < \infty. \end{aligned}$$

5.3 Is Picard iterates right

Till now, we've proved that $\{y_n\}$ converges. However, we still don't know whether $\lim_{n\to\infty} y_n(t)$ is the answer of (29).
Consider $\lim_{n\to\infty} y_n(t) = y$, and we take limit of both sides of:

$$y_{n+1}(t) = y_0 + \int_{t_0}^{t} f(s, y_n(s))\, ds$$

Then we get

$$y = y_0 + \lim_{n\to\infty} \int_{t_0}^{t} f(s, y_n(s))\, ds \tag{34}$$

Then what we need to do is to justify the right hand side equals to $\int_{t_0}^{t} f(s, y(s))$.
One could also understand that the limit could *pass* the integral, as:

$$\lim_{n\to\infty} \int_{t_0}^{t} f(s, y_n(s))\, ds = \int_{t_0}^{t} \left(\lim_{n\to\infty} f(s, y_n(s))\right) ds = \int_{t_0}^{t} f(s, y(s))\, ds$$

We still need to justify this step. For $m > n$, by (33),

$$\begin{aligned}
|y_m(t) - y_n(t)| &\le \sum_{k=n+1}^{m} |y_k(t) - y_{k-1}(t)| \\
&\le \sum_{k=n+1}^{m} \frac{ML^{k-1}(t-t_0)^k}{k!} \\
&\le \sum_{k=n+1}^{m} \frac{ML^{k-1}\alpha^k}{k!}.
\end{aligned}$$

Let $m \to \infty$. Then

$$\sup_{t_0 \le t \le t_0+\alpha} |y(t) - y_n(t)| \le \sum_{k=n+1}^{\infty} \frac{ML^{k-1}\alpha^k}{k!} \longrightarrow 0. \tag{35}$$

Thus,

$$y_n \longrightarrow y \quad \text{uniformly on } [t_0, t_0+\alpha].$$

By the mean value theorem,

$$\begin{aligned}
|f(s, y_n(s)) - f(s, y(s))| &\le L|y_n(s) - y(s)| \\
&\le L \sup_{t_0 \le r \le t_0+\alpha} |y_n(r) - y(r)|.
\end{aligned}$$

Therefore,

$$\begin{aligned}
&\left| \int_{t_0}^{t} f(s, y_n(s))\,ds - \int_{t_0}^{t} f(s, y(s))\,ds \right| \\
&\le \int_{t_0}^{t} |f(s, y_n(s)) - f(s, y(s))|\,ds \\
&\le \alpha L \sup_{t_0 \le r \le t_0+\alpha} |y_n(r) - y(r)| \\
&\longrightarrow 0.
\end{aligned}$$

So the limit can pass through the integral:

$$\lim_{n\to\infty} \int_{t_0}^{t} f(s, y_n(s))\,ds = \int_{t_0}^{t} f(s, y(s))\,ds.$$

Taking the limit in the Picard iteration gives

$$\boxed{y(t) = y_0 + \int_{t_0}^{t} f(s, y(s))\,ds.} \tag{36}$$

We also need to show that $y(t)$ is continuous. First,

$$y_0(t) = y_0 \quad \Longrightarrow \quad y_0 \text{ is continuous.}$$

If y_n is continuous, then

$$s \longmapsto f(s, y_n(s)) \text{ is continuous}$$

and

$$y_{n+1}(t) = y_0 + \int_{t_0}^{t} f(s, y_n(s))\, ds \quad \Longrightarrow \quad y_{n+1} \text{ is continuous.}$$

Thus, every y_n is continuous. Fix $t^* \in [t_0, t_0 + \alpha]$. For every $\varepsilon > 0$, the uniform convergence in (35) gives an N such that

$$\sup_{t_0 \le t \le t_0+\alpha} |y(t) - y_N(t)| < \frac{\varepsilon}{3}.$$

Since y_N is continuous, there exists $\delta > 0$ such that

$$|t - t^*| < \delta \quad \Longrightarrow \quad |y_N(t) - y_N(t^*)| < \frac{\varepsilon}{3}.$$

Therefore,

$$\begin{aligned} |y(t) - y(t^*)| &\le |y(t) - y_N(t)| + |y_N(t) - y_N(t^*)| + |y_N(t^*) - y(t^*)| \\ &< \varepsilon. \end{aligned}$$

Hence,

$$\boxed{y \in C([t_0, t_0 + \alpha]).} \tag{37}$$

Finally,

$$y(t_0) = y_0, \qquad \frac{dy}{dt} = f(t, y(t)).$$

Thus, $y(t)$ is a solution of (29).

5.4 Uniqueness

Theorem 1.5.2. *Let f and $\partial f/\partial y$ be continuous in the rectangle*

$$R: \quad t_0 \le t \le t_0 + a, \quad |y - y_0| \le b.$$

Compute

$$M = \max_{(t,y)\in R} |f(t, y)|, \quad \text{and set } \alpha = \min\left(a, \frac{b}{M}\right).$$

Then, the initial-value problem

$$y' = f(t, y), \quad y(t_0) = y_0 \tag{38}$$

has a unique solution $y(t)$ on the interval $t_0 \le t \le t_0 + \alpha$. In other words, if $y(t)$ and $z(t)$ are two solutions of (38), then $y(t)$ must equal $z(t)$ for $t_0 \le t \le t_0 + \alpha$.

Proof. Theorem 2 guarantees the existence of at least one solution $y(t)$ of (38).

Suppose that $z(t)$ is a second solution of (38). Then,

$$y(t) = y_0 + \int_{t_0}^{t} f(s, y(s))\, ds \quad \text{and} \quad z(t) = y_0 + \int_{t_0}^{t} f(s, z(s))\, ds.$$

Subtracting these two equations gives

$$\begin{aligned} |y(t) - z(t)| &= \left| \int_{t_0}^{t} [f(s, y(s)) - f(s, z(s))]\, ds \right| \\ &\leq \int_{t_0}^{t} |f(s, y(s)) - f(s, z(s))|\, ds \\ &\leq L \int_{t_0}^{t} |y(s) - z(s)|\, ds, \end{aligned}$$

where L is the maximum value of $|\partial f/\partial y|$ for (t, y) in R. As Lemma 5.3 below shows, this inequality implies that $y(t) = z(t)$. Hence, the initial-value problem (38) has a unique solution $y(t)$. □

Lemma 5.3. *Let $w(t)$ be a nonnegative function, with*

$$w(t) \leq L \int_{t_0}^{t} w(s)\, ds.$$

Then, $w(t)$ is identically zero.

Proof. Set

$$U(t) = \int_{t_0}^{t} w(s)\, ds.$$

Then

$$U'(t) = w(t) \leq L \int_{t_0}^{t} w(s)\, ds = LU(t).$$

Therefore,

$$\begin{aligned} \frac{d}{dt}\left(e^{-L(t-t_0)} U(t)\right) &= e^{-L(t-t_0)}\left(U'(t) - LU(t)\right) \\ &\leq 0. \end{aligned}$$

Thus,

$$0 \leq e^{-L(t-t_0)} U(t) \leq U(t_0) = 0,$$

so $U(t) = 0$. Finally,

$$0 \leq w(t) \leq L \int_{t_0}^{t} w(s)\, ds = LU(t) = 0.$$

Hence, $\boxed{w(t) = 0}$ for $t_0 \leq t \leq t_0 + \alpha$. □

5.5 Proof of (32)

First, consider a sequence $\{a_n\}_{n\geq 1} \subset \mathbb{R}$ such that

$$\sum_{n=1}^{\infty} |a_n| < \infty.$$

Define

$$S_n = \sum_{k=1}^{n} a_k, \qquad A_n = \sum_{k=1}^{n} |a_k|.$$

Since $\{A_n\}$ converges, it is a Cauchy sequence. Thus,

$$\forall \varepsilon > 0, \qquad \exists N \in \mathbb{N}, \qquad m > n \geq N$$

$$\Downarrow$$

$$|A_m - A_n| = \sum_{k=n+1}^{m} |a_k| < \varepsilon.$$

Then

$$\begin{aligned} |S_m - S_n| &= \left| \sum_{k=n+1}^{m} a_k \right| \\ &\leq \sum_{k=n+1}^{m} |a_k| \\ &< \varepsilon. \end{aligned}$$

Therefore,

$$\{S_n\} \text{ is Cauchy} \overset{\mathbb{R} \text{ is complete}}{\Longrightarrow} S_n \longrightarrow S \in \mathbb{R},$$

and hence

$$\boxed{\sum_{n=1}^{\infty} |a_n| < \infty \quad \Longrightarrow \quad \sum_{n=1}^{\infty} a_n \text{ converges.}} \tag{39}$$

Chapter 2

Second Order Linear Differential Equations

1 Introduction

We've considered first order differential equations. Now we try to understand second order.
Consider

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = f(t, y, \frac{\mathrm{d}y}{\mathrm{d}t}), \quad y(t_0) = y_0, y'(t_0) = y'_0 \tag{1}$$

Think about how we define linear first order equation before. Now we define second order linear differential equation.

Definition 2.1.1 (Second order linear differential equation)**.**
The equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + p(t)\frac{\mathrm{d}y}{\mathrm{d}t} + q(t)y = g(t) \tag{2}$$

is called second order linear differential equation.

Sadly, almost all second order differential equations are extremely difficult to solve. Till now we can only solve (2)

Consider an easier case of (2) when $g(t) = 0$. Like what we did in forst order, we can also define second order linear homogeneous equation.

Definition 2.1.2 (Second order linear homogeneous equation)**.**
The equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + p(t)\frac{\mathrm{d}y}{\mathrm{d}t} + q(t)y = 0 \tag{3}$$

is called second order linear homogeneous equation.

However, it could be very difficult even in solving second order linear homogeneous equation. We'll first consider whether it actually has a solution (or solutions) and its properties.

2 Algebraic properties of solutions

Consider

$$\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + p(t)\frac{\mathrm{d}y}{\mathrm{d}t} + q(t)y = 0 \tag{4}$$

Or better, consider initial value problem:

$$\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + p(t)\frac{\mathrm{d}y}{\mathrm{d}t} + q(t)y = 0, \quad y_0 = y(t_0), y_0' = y(t_0) \tag{5}$$

Then we consider an operator $L[y](t)$:

$$L[y](t) = \frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + p(t)\frac{\mathrm{d}y}{\mathrm{d}t} + q(t)y \tag{6}$$

One can understand $L[y](t)$ as 'function on functions'. We input a function $y(t)$, then output a function $L[y](t)$. With the help of operator, we could analysis the solutions of (4) easier.

Property 1 $L[cy] = cL[y]$
Property 2 $L[y_1 + y_2] = L[y_1] + L[y_2]$

Definition 2.2.1 (Linear operator)**.**
An operator L which satisfies properties 1 and 2 is called a linear operator. All other operators are non-linear.

Now we give a theorem which will be proved very later:

Theorem 2.2.1 (Existence–uniqueness Theorem)**.** *Let the functions $p(t)$ and $q(t)$ be continuous in the open interval $\alpha < t < \beta$. Then, there exists one, and only one function $y(t)$ satisfying the differential equation (4) on the entire interval $\alpha < t < \beta$, and the prescribed initial conditions*

$$y(t_0) = y_0, \qquad y'(t_0) = y_0'.$$

In particular, any solution $y = y(t)$ of (3) which satisfies $y(t_0) = 0$ and $y'(t_0) = 0$ at some time $t = t_0$ must be identically zero.

Thus the solution of (4) could be written in the form of (6) as

$$L[y](t) = 0 \tag{7}$$

Note the right hand side is also a function instead a scalar.

For any solution $y(t)$ of (7), $cy(t)$ is also the solution. For any solutions $y_1(t)$, $y_2(t)$ of (7), $c_1y_1(t) + c_2y_2(t)$ is also the solution.

Assume we already obtain two solutions y_1 and y_2 of (5), and $y(t)$ is any solution of (5). We must find constants c_1 and c_2 such that $y(t) = c_1y_1(t) + c_2y_2(t)$. The constants c_1 and c_2, if they exist, must satisfy the two equations

$$c_1 y_1(t_0) + c_2 y_2(t_0) = y_0,$$

$$c_1 y_1'(t_0) + c_2 y_2'(t_0) = y_0'.$$

Then it gives

$$c_1 \big[y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0)\big] = y_0 y_2'(t_0) - y_0' y_2(t_0),$$

$$c_2 \big[y_1'(t_0)y_2(t_0) - y_1(t_0)y_2'(t_0)\big] = y_0 y_1'(t_0) - y_0' y_1(t_0).$$

Hence,

$$c_1 = \frac{y_0 y_2'(t_0) - y_0' y_2(t_0)}{y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0)},$$

$$c_2 = \frac{y_0' y_1(t_0) - y_0 y_1'(t_0)}{y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0)}$$

if $y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0) \neq 0$.
Then, we come to **Theorem 2.2.2**.

Theorem 2.2.2. *Let $y_1(t)$ and $y_2(t)$ be two solutions of (4) on the interval $\alpha < t < \beta$, with*

$$y_1(t)y_2'(t) - y_1'(t)y_2(t)$$

unequal to zero in this interval. Then,

$$y(t) = c_1 y_1(t) + c_2 y_2(t)$$

is the general solution of (4).

Above we've already showed the idea of the proof. The entire proof will be in 2.1.

Definition 2.2.2 (Wronskian)**.**
The quantity

$$y_1(t)y_2'(t) - y_1'(t)y_2(t)$$

is called the Wronskian *of y_1 and y_2, and is denoted by*

$$W(t) = W[y_1, y_2](t).$$

One could consider a matrix form:

$$\begin{bmatrix} y_1 & y_2 \\ y_1' & y_2' \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} c_1 y_1 + c_2 y_2 \\ c_1 y_1' + c_2 y_2' \end{bmatrix} = \begin{bmatrix} y \\ y' \end{bmatrix} \tag{8}$$

Then we come to Theorem 2.2.3 which shows some property of Wronskian, and we would prove Theorem 2.2.3 with the aid of Lemma 2.4.

Theorem 2.2.3. *Let $p(t)$ and $q(t)$ be continuous in the interval $\alpha < t < \beta$, and let $y_1(t)$ and $y_2(t)$ be two solutions of (3). Then, $W[y_1, y_2](t)$ is either identically zero, or is never zero, on the interval $\alpha < t < \beta$.*

Lemma 2.4. *Let $y_1(t)$ and $y_2(t)$ be two solutions of the linear differential equation*

$$y'' + p(t)y' + q(t)y = 0.$$

Then, their Wronskian

$$W(t) = W[y_1, y_2](t) = y_1(t)y_2'(t) - y_1'(t)y_2(t)$$

satisfies the first-order differential equation

$$W' + p(t)W = 0.$$

Entire proof will be in 2.2.

Easy to see, if $y_1 = cy_2$, then $W[y_1, y_2](t) = 0$. Also, it could be proved that if $W[y_1, y_2](t) = 0$, then $y_1 = cy_2$.

Theorem 2.2.5. *Let $y_1(t)$ and $y_2(t)$ be two solutions of (3) on the interval $\alpha < t < \beta$, and suppose that $W[y_1, y_2](t_0) = 0$ for some t_0 in this interval. Then, one of these solutions is a constant multiple of the other.*

Entire proof will be in 2.3.

Definition 2.2.3 (Linear dependent)**.**
The functions $y_1(t)$ and $y_2(t)$ are said to be linearly dependent *on an interval I if one of these functions is a constant multiple of the other on I. The functions $y_1(t)$ and $y_2(t)$ are said to be* linearly independent *on an interval I if they are not linearly dependent on I.*

Lemma 2.6 (to Theorem 2.2.5)**.** *Two solutions $y_1(t)$ and $y_2(t)$ of (3) are linearly independent on the interval $\alpha < t < \beta$ if, and only if, their Wronskian is not equal to zero on this interval. Thus, two solutions $y_1(t)$ and $y_2(t)$ form a fundamental set of solutions of (3) on the interval $\alpha < t < \beta$ if, and only if, they are linearly independent on this interval.*

2.1 Proof of Theorem 2.2.2

Theorem 2.2.2. *Let $y_1(t)$ and $y_2(t)$ be two solutions of (4) on the interval $\alpha < t < \beta$, with*

$$y_1(t)y_2'(t) - y_1'(t)y_2(t)$$

unequal to zero in this interval. Then,

$$y(t) = c_1y_1(t) + c_2y_2(t)$$

is the general solution of (4).

Proof. Let $y(t)$ be any solution of (4). We must find constants c_1 and c_2 such that $y(t) = c_1y_1(t) + c_2y_2(t)$. To this end, pick a time t_0 in the interval (α, β) and let y_0 and y_0' denote the values of y and y' at $t = t_0$. The constants c_1 and c_2, if they exist, must satisfy the two equations

$$c_1y_1(t_0) + c_2y_2(t_0) = y_0,$$

$$c_1y_1'(t_0) + c_2y_2'(t_0) = y_0'.$$

Multiplying the first equation by $y_2'(t_0)$, the second equation by $y_2(t_0)$, and subtracting gives

$$c_1\left[y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0)\right] = y_0y_2'(t_0) - y_0'y_2(t_0).$$

Similarly, multiplying the first equation by $y_1'(t_0)$, the second equation by $y_1(t_0)$, and subtracting gives

$$c_2\left[y_1'(t_0)y_2(t_0) - y_1(t_0)y_2'(t_0)\right] = y_0y_1'(t_0) - y_0'y_1(t_0).$$

Hence,

$$c_1 = \frac{y_0y_2'(t_0) - y_0'y_2(t_0)}{y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0)}$$

and

$$c_2 = \frac{y_0'y_1(t_0) - y_0y_1'(t_0)}{y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0)}$$

if $y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0) \neq 0$.
Now, let

$$\phi(t) = c_1y_1(t) + c_2y_2(t)$$

for this choice of constants c_1, c_2. We know that $\phi(t)$ satisfies (4), since it is a linear combination of solutions of (4). Moreover, by construction, $\phi(t_0) = y_0$ and $\phi'(t_0) = y_0'$. Thus, $y(t)$ and $\phi(t)$ satisfy the same second-order linear homogeneous equation and the same initial conditions. Therefore, by the uniqueness part of Theorem 2.2.1, $y(t)$ must be identically equal to $\phi(t)$; that is,

$$y(t) = c_1y_1(t) + c_2y_2(t), \quad \alpha < t < \beta.$$

Theorem 2 is an extremely useful theorem since it reduces the problem of finding all solutions of (4), of which there are infinitely many, to the much simpler problem of finding just two solutions $y_1(t), y_2(t)$. The only condition imposed on the solutions

$y_1(t)$ and $y_2(t)$ is that the quantity

$$y_1(t)y_2'(t) - y_1'(t)y_2(t)$$

be unequal to zero for $\alpha < t < \beta$. When this is the case, we say that $y_1(t)$ and $y_2(t)$ are a *fundamental set of solutions* of (4), since all other solutions of (4) can be obtained by taking linear combinations of $y_1(t)$ and $y_2(t)$. $\square$

2.2 Proof of Theorem 2.2.3 and Lemma 2.4

Lemma 2.4. *Let $y_1(t)$ and $y_2(t)$ be two solutions of the linear differential equation*

$$y'' + p(t)y' + q(t)y = 0.$$

Then, their Wronskian

$$W(t) = W[y_1, y_2](t) = y_1(t)y_2'(t) - y_1'(t)y_2(t)$$

satisfies the first-order differential equation

$$W' + p(t)W = 0.$$

Proof. Observe that

$$W'(t) = \frac{d}{dt}(y_1y_2' - y_1'y_2) = y_1y_2'' + y_1'y_2' - y_1'y_2' - y_1''y_2 = y_1y_2'' - y_1''y_2.$$

Since y_1 and y_2 are solutions of $y'' + p(t)y' + q(t)y = 0$, we know that

$$y_2'' = -p(t)y_2' - q(t)y_2$$

and

$$y_1'' = -p(t)y_1' - q(t)y_1.$$

Hence,

$$W'(t) = y_1\left[-p(t)y_2' - q(t)y_2\right] - y_2\left[-p(t)y_1' - q(t)y_1\right] = -p(t)\left[y_1y_2' - y_1'y_2\right] = -p(t)W(t).$$

$\square$

Theorem 2.2.3. *Let $p(t)$ and $q(t)$ be continuous in the interval $\alpha < t < \beta$, and let $y_1(t)$ and $y_2(t)$ be two solutions of (3). Then, $W[y_1, y_2](t)$ is either identically zero, or is never zero, on the interval $\alpha < t < \beta$.*

Proof of Theorem 3. Pick any t_0 in the interval (α, β). From Lemma 1,

$$W[y_1, y_2](t) = W[y_1, y_2](t_0)\exp\left(-\int_{t_0}^{t} p(s)\,ds\right).$$

Now, $\exp\left(-\int_{t_0}^{t} p(s)\,ds\right)$ is unequal to zero for $\alpha < t < \beta$. Therefore, $W[y_1, y_2](t)$ is either identically zero, or is never zero. $\square$

2.3 Proof of Theorem 2.2.5

Theorem 2.2.5. *Let $y_1(t)$ and $y_2(t)$ be two solutions of (3) on the interval $\alpha < t < \beta$, and suppose that $W[y_1, y_2](t_0) = 0$ for some t_0 in this interval. Then, one of these solutions is a constant multiple of the other.*

Here we provide two method to prove:

Proof #1. Suppose that $W[y_1, y_2](t_0) = 0$. Then, recall the matrix form, the equations

$$c_1 y_1(t_0) + c_2 y_2(t_0) = 0,$$
$$c_1 y_1'(t_0) + c_2 y_2'(t_0) = 0$$

have a nontrivial solution c_1, c_2; that is, a solution c_1, c_2 with $|c_1| + |c_2| \neq 0$. Let $y(t) = c_1 y_1(t) + c_2 y_2(t)$, for this choice of constants c_1, c_2. We know that $y(t)$ is a solution of (3), since it is a linear combination of $y_1(t)$ and $y_2(t)$. Moreover, by construction, $y(t_0) = 0$ and $y'(t_0) = 0$. Therefore, by Theorem 2.2.1, $y(t)$ is identically zero, so that

$$c_1 y_1(t) + c_2 y_2(t) = 0, \quad \alpha < t < \beta.$$

If $c_1 \neq 0$, then $y_1(t) = -(c_2/c_1) y_2(t)$, and if $c_2 \neq 0$, then $y_2(t) = -(c_1/c_2) y_1(t)$. In either case, one of these solutions is a constant multiple of the other. □ □

Proof #2. Suppose that $W[y_1, y_2](t_0) = 0$. Then, by Theorem 2.2.3, $W[y_1, y_2](t)$ is identically zero. Assume that $y_1(t) y_2(t) \neq 0$ for $\alpha < t < \beta$. Then, dividing both sides of the equation

$$y_1(t) y_2'(t) - y_1'(t) y_2(t) = 0$$

by $y_1(t) y_2(t)$ gives

$$\frac{y_2'(t)}{y_2(t)} - \frac{y_1'(t)}{y_1(t)} = 0.$$

This equation implies that $y_1(t) = c y_2(t)$ for some constant c.
Next, suppose that $y_1(t) y_2(t)$ is zero at some point $t = t^*$ in the interval $\alpha < t < \beta$. Without loss of generality, we may assume that $y_1(t^*) = 0$, since otherwise we can relabel y_1 and y_2. In this case it is simple to show that either $y_1(t) \equiv 0$, or

$$y_2(t) = \frac{y_2'(t^*)}{y_1'(t^*)} y_1(t).$$

This completes the proof of Theorem 2.2.5. □ □

3 Linear differential equations with constant coefficients

Consider
$$L[y](t) = a\frac{\mathrm{d}^2 y(t)}{\mathrm{d}t^2} + b\frac{\mathrm{d}y(t)}{\mathrm{d}t} + cy(t) = 0, \qquad a \neq 0, \tag{9}$$
where a, b, c are constants.
By Theorem 2.2.2, we know that we only need to find two linearly independent solutions of (9). However, the theorem does not tell us how to find them.
Easy to see, that y'', y', y should be the same type. One natural idea is to try:
$$y(t) = e^{rt}.$$
Then
$$\begin{aligned} L[e^{rt}] &= a\frac{d^2}{dt^2}e^{rt} + b\frac{d}{dt}e^{rt} + ce^{rt} \\ &= ar^2 e^{rt} + bre^{rt} + ce^{rt} \\ &= (ar^2 + br + c)e^{rt}. \end{aligned}$$
Since $e^{rt} \neq 0$,
$$L[e^{rt}] = 0 \quad \Longleftrightarrow \quad ar^2 + br + c = 0.$$

Definition 2.3.1 (Characteristic equation)**.**
The equation
$$ar^2 + br + c = 0 \tag{10}$$
is called the characteristic equation of (9).

The roots are
$$r_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \qquad r_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}. \tag{11}$$
Thus, the form of the solution depends on
$$\Delta = b^2 - 4ac.$$
We consider the three cases
$$\Delta > 0, \qquad \Delta < 0, \qquad \Delta = 0.$$

3.1 Real and distinct roots

Suppose
$$\Delta = b^2 - 4ac > 0.$$
Then r_1, r_2 in (11) are real and $r_1 \neq r_2$. Therefore,
$$y_1(t) = e^{r_1 t}, \qquad y_2(t) = e^{r_2 t}$$

are two solutions of (9).
Very easy to see, y_1 is not a constant multiple of y_2.
Thus, y_1, y_2 form a fundamental set of solutions. By Theorem 2.2.2,

$$\boxed{y(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}.} \tag{12}$$

3.2 Complex roots

There are two solutions:

$$r_1 = \frac{-b + i\sqrt{-b^2 + 4ac}}{2a}, \qquad r_2 = \frac{-b - i\sqrt{-b^2 + 4ac}}{2a}. \tag{13}$$

We use two notions:

$$\alpha = -\frac{b}{2a}, \quad \beta = \frac{\sqrt{4ac - b^2}}{2a}$$

However, there are two problem remaining to be solved:
Problem 1 e^{rt} has not been well-defined yet for complex r.
Problem 2 Even e^{rt} is well-defined, we still need to find the real-value solution.
First, we try to solve **Problem 1**.

Lemma 3.1. *Suppose that $a, b, c \in \mathbb{R}$ and*

$$y(t) = u(t) + iv(t)$$

is a complex-valued solution of (9). Then $u(t)$ and $v(t)$ are real-valued solutions of (9).

Proof. Since $L[y] = 0$,

$$\begin{aligned} 0 &= L[u + iv] \\ &= L[u] + iL[v] \\ &= \big(au'' + bu' + cu\big) + i\big(av'' + bv' + cv\big). \end{aligned}$$

Thus,

$$au'' + bu' + cu = 0, \qquad av'' + bv' + cv = 0.$$

Hence, both u and v solve (9). □

Then, we try to solve **Problem 2**. Now we write $e^{(\alpha+i\beta)t}$ explicitly. From the power series,

$$\begin{aligned} e^{i\beta t} &= 1 + i\beta t + \frac{(i\beta t)^2}{2!} + \frac{(i\beta t)^3}{3!} + \cdots \\ &= \left(1 - \frac{\beta^2 t^2}{2!} + \frac{\beta^4 t^4}{4!} - \cdots\right) i \left(\beta t - \frac{\beta^3 t^3}{3!} + \frac{\beta^5 t^5}{5!} - \cdots\right) \\ &= \cos(\beta t) + i\sin(\beta t). \end{aligned}$$

Therefore,

$$e^{(\alpha+i\beta)t} = e^{\alpha t}\big(\cos(\beta t) + i\sin(\beta t)\big). \tag{14}$$

Also,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}e^{(\alpha+i\beta)t} &= e^{\alpha t}\left[\alpha\cos(\beta t) - \beta\sin(\beta t) + i\big(\alpha\sin(\beta t) + \beta\cos(\beta t)\big)\right] \\ &= (\alpha+i\beta)e^{(\alpha+i\beta)t}.\end{aligned}$$

Since $r_1 = \alpha + i\beta$ satisfies the characteristic equation, $e^{r_1 t}$ is a complex-valued solution of (9). By Lemma 3.1, its real and imaginary parts

$$y_1(t) = e^{\alpha t}\cos(\beta t), \qquad y_2(t) = e^{\alpha t}\sin(\beta t)$$

are real-valued solutions.
Also,

$$\begin{aligned}W[y_1,y_2](t) &= \begin{vmatrix} e^{\alpha t}\cos(\beta t) & e^{\alpha t}\sin(\beta t) \\ e^{\alpha t}\big(\alpha\cos(\beta t) - \beta\sin(\beta t)\big) & e^{\alpha t}\big(\alpha\sin(\beta t) + \beta\cos(\beta t)\big)\end{vmatrix} \\ &= \beta e^{2\alpha t} \\ &\neq 0.\end{aligned}$$

Thus,

$$\boxed{y(t) = e^{\alpha t}\left[c_1\cos(\beta t) + c_2\sin(\beta t)\right].} \tag{15}$$

Remark 3.2. One would ask that we use $e^{r_1 t}$ to get two solutions

$$y_1(t) = e^{\alpha t}\cos(\beta t), \qquad y_2(t) = e^{\alpha t}\sin(\beta t)$$

Then if we use $e^{r_2 t}$, what can we get?

$$e^{r_2 t} = e^{\alpha t}[\cos\beta t - i\sin\beta t]$$

3.3 Equal roots, and reduction of order

Suppose

$$\Delta = b^2 - 4ac = 0.$$

So

$$r = -\frac{b}{2a}, \quad y_1(t) = e^{-bt/(2a)}.$$

However, we still need to get another solution. We would write it as

$$y_2(t) = y_1(t)v(t) \tag{16}$$

We would first consider a general case as (17). We need to plug (16) back to (17):

$$y'' + p(t)y' + q(t)y = 0. \tag{17}$$

Then we find the derivative of y_2:

$$y_2' = y_1'v + y_1v', \qquad y_2'' = y_1''v + 2y_1'v' + y_1v''.$$

Substituting into (17) gives

$$\begin{aligned}0 &= \big(y_1'' + py_1' + qy_1\big)v + \big(2y_1' + py_1\big)v' + y_1v'' \\ &= \big(2y_1' + py_1\big)v' + y_1v'',\end{aligned}$$

Hence, $y_2(t) = y_1(t)v(t)$ is a solution if v satisfies:

$$y_1\frac{\mathrm{d}^2v}{\mathrm{d}t^2} + \left[2\frac{\mathrm{d}y_1}{\mathrm{d}t} + p(t)y_1\right]\frac{\mathrm{d}v}{\mathrm{d}t} = 0 \tag{18}$$

Note that (18) is a first-order linear equation. So:

$$\frac{\mathrm{d}v}{\mathrm{d}t} = c\exp\Big[-\int\big[2\frac{y_1'(t)}{y_1(t)} + p(t)\big]dt\Big] = c\frac{\exp\big(-\int p(t)\,dt\big)}{y_1^2(t)}. \tag{19}$$

Only one solution is needed, so we could just set $c = 1$:

$$\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\exp\big(-\int p(t)\,dt\big)}{y_1^2(t)}. \tag{20}$$

$$\Downarrow$$

$$v(t) = \int\frac{\exp\big(-\int p(t)\,dt\big)}{y_1^2(t)}\,dt.$$

Therefore, a second solution is

$$\boxed{y_2(t) = y_1(t)\int\frac{\exp\big(-\int p(t)\,dt\big)}{y_1^2(t)}\,dt.} \tag{21}$$

This method changes the problem from a second-order equation to a first-order equation, so it is called *reduction of order*.

Now return to (9). Dividing it by a, we obtain

$$y'' + \frac{b}{a}y' + \frac{c}{a}y = 0.$$

Thus,

$$p(t) = \frac{b}{a}, \qquad y_1(t) = e^{-bt/(2a)}.$$

The integrand in (21) becomes

$$\begin{aligned}\frac{\exp\big(-\int p(t)\,dt\big)}{y_1^2(t)} &= \frac{e^{-bt/a}}{\big(e^{-bt/(2a)}\big)^2} \\ &= 1.\end{aligned}$$

Therefore,

$$v(t) = \int 1\,dt = t, \qquad y_2(t) = te^{-bt/(2a)}.$$

Their Wronskian is

$$W[y_1, y_2](t) = e^{-bt/a} \neq 0.$$

Hence, the general solution for equal roots is

$$\boxed{y(t) = c_1 e^{-bt/(2a)} + c_2 t e^{-bt/(2a)} = (c_1 + c_2 t)\, e^{-bt/(2a)}.} \tag{22}$$

4 The nonhomogeneous equation

Now consider

$$L[y](t) = \frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + p(t)\frac{\mathrm{d}y}{\mathrm{d}t} + q(t)y = g(t), \tag{23}$$

where $p(t)$, $q(t)$, and $g(t)$ are continuous on $\alpha < t < \beta$.

Definition 2.4.1 (Particular solution)**.**
Any one function $\psi(t)$ satisfying

$$L[\psi](t) = g(t)$$

is called a particular solution of (23).

Also, we may consider a very *look* $-$ *like* homogeneous equation:

$$L[y](t) = \frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + p(t)\frac{\mathrm{d}y}{\mathrm{d}t} + q(t)y = 0 \tag{24}$$

Consider solutions of (24): $y_1(t)$ and $y_2(t)$. Easy to see, for (23), $\forall c_1, c_2$:

$$\begin{aligned} L[c_1y_1 + c_2y_2 + \phi](t) &= L[c_1y_1](t) + L[c_2y_2](t) + L[\phi](t) \\ &= c_1L[y_1](t) + c_2L[y_2](t) + L[\phi](t) \\ &= L[\phi](t) \\ &= g(t) \end{aligned}$$

One may get some understanding of the connection of the solutions of (23) and (24).

Lemma 4.1. *The difference of any two solutions of (23) is a solution of (24).*

The proof shall be very easy as one gets to understand above.

Theorem 2.4.2. *Suppose that $y_1(t)$ and $y_2(t)$ are two linearly independent solutions of (24), and $\psi(t)$ is one particular solution of (23). Every solution of (23) is of the form*

$$\boxed{y(t) = c_1y_1(t) + c_2y_2(t) + \psi(t).} \tag{25}$$

Proof. Let $y(t)$ be any solution of (23). By Lemma 4.1,

$$\phi(t) = y(t) - \psi(t)$$

is a solution of (24). Since y_1, y_2 are linearly independent, Theorem 2.2.2 gives

$$\phi(t) = c_1y_1(t) + c_2y_2(t).$$

Therefore,

$$y(t) = c_1y_1(t) + c_2y_2(t) + \psi(t).$$

Conversely,

$$\begin{aligned} L[c_1y_1 + c_2y_2 + \psi](t) &= c_1L[y_1](t) + c_2L[y_2](t) + L[\psi](t) \\ &= 0 + 0 + g(t) \\ &= g(t). \end{aligned}$$

Thus, every function in (25) is also a solution of (23). $\square$

5 Series solutions

MAP

In the main part of 5.1 *Series as solutions*, we will introduce series and its basic properties.

$\Downarrow$

In 5.3 *Euler equations*, we will analysis a simple equation $t^2\frac{\mathrm{d}^2y}{\mathrm{d}t^2}+\alpha t\frac{\mathrm{d}y}{\mathrm{d}t}+\beta y=0$ and many cases of its roots.

$\Downarrow$

In 5.2 *Singular points, regular singular points* we will try to analysis $L[y]=P(t)\frac{\mathrm{d}^2y}{\mathrm{d}t^2}+Q(t)\frac{\mathrm{d}y}{\mathrm{d}t}+R(t)y=0$ near t^* as $P(t^*)=0$

$\Downarrow$

In 5.4 *The Method of Frobenius* we will introduce when we can solve $L[y]=\frac{\mathrm{d}^2y}{\mathrm{d}t^2}+p(t)\frac{\mathrm{d}y}{\mathrm{d}t}+q(t)y=0$ and how.

5.1 Series as solutions

Consider the homogeneous linear equation

$$L[y]=P(t)\frac{\mathrm{d}^2y}{\mathrm{d}t^2}+Q(t)\frac{\mathrm{d}y}{\mathrm{d}t}+R(t)y=0.$$

with $P(t)$ unequal to 0 in the interval $\alpha<t<\beta$.

If P,Q,R are constant, we've also talked about it in 3.
However, if any of P,Q,R is just a bit complex, we may not be able to solve them directly. Instead, we may use a polynomial (finite degree or infinite degree) to write the solution:

$$y(t)=a_0+a_1t+\cdots+a_Nt^N+\cdots$$

Or we may even write the solution in a more complex polynomial:

$$y(t)=t^r\sum_{n=0}^{\infty}a_nt^n$$

One could view *Differential Equations and Their Applications* P186, Example 1 to get a better understanding.

Before we discuss deeper, we may now get some knowledge about power series.

1. An infinite series

$$y(t)=a_0+a_1(t-t_0)+a_2(t-t_0)^2+\cdots=\sum_{n=0}^{\infty}a_n(t-t_0)^n \tag{26}$$

is called a power series about $t=t_0$.

2. All power series have an interval of convergence. This means that there exists a nonnegative number ρ such that the infinite series (26) converges for $|t-t_0| < \rho$, and diverges for $|t-t_0| > \rho$. The number ρ is called the radius of convergence of the power series.

3. The power series (26) can be differentiated and integrated term by term, and the resultant series have the same interval of convergence.

4. The simplest method (if it works) for determining the interval of convergence of the power series (26) is the Cauchy ratio test. Suppose that the absolute value of a_{n+1}/a_n approaches a limit λ as n approaches infinity. Then, the power series (26) converges for $|t - t_0| < 1/\lambda$, and diverges for $|t - t_0| > 1/\lambda$.

5. The product of two power series $\sum_{n=0}^{\infty} a_n(t-t_0)^n$ and $\sum_{n=0}^{\infty} b_n(t-t_0)^n$ is again a power series of the form $\sum_{n=0}^{\infty} c_n(t-t_0)^n$, with

$$c_n = a_0b_n + a_1b_{n-1} + \cdots + a_nb_0.$$

The quotient

$$\frac{a_0 + a_1t + a_2t^2 + \cdots}{b_0 + b_1t + b_2t^2 + \cdots}$$

of two power series is again a power series, provided that $b_0 \neq 0$.

6. Many of the functions $f(t)$ that arise in applications can be expanded in power series; that is, we can find coefficients $a_0, a_1, a_2, \ldots$ so that

$$f(t) = a_0 + a_1(t-t_0) + a_2(t-t_0)^2 + \cdots = \sum_{n=0}^{\infty} a_n(t-t_0)^n \tag{27}$$

Such functions are said to be *analytic* at $t = t_0$, and the series (27) is called the Taylor series of f about $t = t_0$. It can easily be shown that if f admits such an expansion, then, of necessity,

$$a_n = \frac{f^{(n)}(t_0)}{n!},$$

where $f^{(n)}(t) = d^nf(t)/dt^n$.

7. The interval of convergence of the Taylor series of a function $f(t)$, about t_0, can be determined directly through the Cauchy ratio test and other similar methods, or indirectly, through the following theorem of complex analysis.

Theorem 2.5.1. *Let the variable t assume complex values, and let z_0 be the point closest to t_0 at which f or one of its derivatives fails to exist. Compute the distance ρ, in the complex plane, between t_0 and z_0. Then, the Taylor series of f about t_0 converges for $|t - t_0| < \rho$, and diverges for $|t - t_0| > \rho$.*

Example 2.5.1 (Radius of convergence)**.**
One great example is the following:

$$\frac{1}{1+t^2} = 1 - t^2 + t^4 - t^6 + \cdots .$$

$1/(1+t^2)$ *is defined for every real* t*, but it will goes to infity at*

$$t = \pm i.$$

The distance from 0 *to either* $\pm i$ *is* 1*, so the series converges only for*

$$|t| < 1.$$

Consider a very general differential equation:

$$P(t)\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + Q(t)\frac{\mathrm{d}y}{\mathrm{d}t} + R(t)y = 0. \tag{28}$$

where $P(t), Q(t), R(t)$ are power series about t_0, which means:

$$P(t) = P_0 + P_1(t - t_0) + \cdots$$

$$Q(t) = Q_0 + Q_1(t - t_0) + \cdots$$

$$R(t) = R_0 + R_1(t - t_0) + \cdots$$

And suppose

$$y(t) = a_0 + a_1(t - t_0) + \cdots$$

Theorem 2.5.2. *Suppose*

$$\frac{Q(t)}{P(t)} \quad \text{and} \quad \frac{R(t)}{P(t)}$$

have Taylor series centered at t_0 *which converge for* $|t - t_0| < \rho$*. Then every solution of*

$$P(t)\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + Q(t)\frac{\mathrm{d}y}{\mathrm{d}t} + R(t)y = 0 \tag{29}$$

is analytic at t_0*. Its Taylor series converges at least for* $|t - t_0| < \rho$*. The coefficients* $a_2, a_3, \ldots,$ *in the Taylor series expansion*

$$y(t) = a_0 + a_1(t - t_0) + a_2(t - t_0)^2 + \cdots \tag{30}$$

are determined by plugging the series (30) into the differential equation (29) and setting the sum of the coefficients of like powers of t *in this expression equal to zero.*

Remark 5.3. The interval of convergence of the Taylor series expansion of any solution $y(t)$ of (28) is determined, usually, by the interval of convergence of the power series $Q(t)/P(t)$ and $R(t)/P(t)$, rather than by the interval of convergence of the power series $P(t), Q(t)$, and $R(t)$. This is because the differential equation (28) must be put in the standard form

$$\frac{d^2 y}{dt^2} + p(t)\frac{dy}{dt} + q(t)y = 0$$

whenever we examine questions of existence and uniqueness.

Remark 5.4. In Theorem 2.5.2 we mentioned '*setting the sum of the coefficients of like powers of t in this expression equal to zero*'. It means if we plug the series (30) into the differential equation (29), we could get a equation with the form:

$$L[y](t) = \sum_{n=0}^{\infty} g(n, a_0, a_1, a_2, \cdots)t^n \quad = 0$$

Then we set $g(n, a_0, a_1, a_2, \cdots) = 0$, and we can solve $a_0, a_1, \cdots$

5.2 Singular points, regular singular points

Definition 2.5.1 (Singular point).
The differential equation

$$L[y] = P(t)\frac{d^2y}{dt^2} + Q(t)\frac{dy}{dt} + R(t)y = 0 \tag{31}$$

is said to be singular at $t = t_0$ *if* $P(t_0) = 0$.

Solutions of (31) frequently become unstable in a neighborhood of the singular point t_0.
Our goal is to find a class of singular equations which we can solve for t near t_0. The most famous one could be seen in 5.3 *Euler equations*:

$$t^2\frac{d^2y}{dt^2} + \alpha t\frac{dy}{dt} + \beta y = 0 \tag{32}$$

Our goal, which will be shown in 5.4 *the Method of Frobenius*, is to find a class of singular differential equations which is more general than the Euler equation
To this end we rewrite (32) in the form

$$\frac{d^2y}{dt^2} + \frac{\alpha}{t}\frac{dy}{dt} + \frac{\beta}{t^2}y = 0. \tag{33}$$

A very natural generalization of (33) is the equation

$$L[y] = \frac{d^2y}{dt^2} + p(t)\frac{dy}{dt} + q(t)y = 0 \tag{34}$$

where $p(t)$ and $q(t)$ can be expanded in series of the form

$$p(t) = \frac{p_0}{t} + p_1 + p_2t + p_3t^2 + \cdots \tag{35}$$

$$q(t) = \frac{q_0}{t^2} + \frac{q_1}{t} + q_2 + q_3t + q_4t^2 + \cdots \tag{36}$$

Definition 2.5.2 (regular singular point).
The equation (34) is said to have a regular singular point *at* $t = 0$ *if* $p(t)$ *and* $q(t)$ *have series expansions of the form (35) and (36). Equivalently,* $t = 0$ *is a regular singular point of (34) if the functions* $tp(t)$ *and* $t^2q(t)$ *are analytic at*

$t = 0$. Equation (34) is said to have a regular singular point at $t = t_0$ if the functions $(t - t_0)p(t)$ and $(t - t_0)^2 q(t)$ are analytic at $t = t_0$. A singular point of (34) which is not regular is called irregular.

5.3 Euler equations

Definition 2.5.3 (Euler equation)**.**
The equation

$$t^2 \frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + \alpha t \frac{\mathrm{d}y}{\mathrm{d}t} + \beta y = 0$$

is called an Euler equation.

Observing that

$$t^2 \leftrightarrow \frac{\mathrm{d}^2 y}{\mathrm{d}t^2}, \quad t \leftrightarrow \frac{\mathrm{d}y}{\mathrm{d}t}$$

One natural idea is to try $y = t^r$,

$$\begin{aligned} L[t^r] &= t^2 r(r-1)t^{r-2} + \alpha t r t^{r-1} + \beta t^r \\ &= [r(r-1) + \alpha r + \beta]\, t^r. \end{aligned}$$

Set

$$F(r) = r(r-1) + \alpha r + \beta = r^2 + (\alpha - 1)r + \beta.$$

Since $t^r \neq 0$ for $t > 0$,

$$L[t^r] = 0 \quad \Longleftrightarrow \quad \boxed{F(r) = 0}.$$

There are three cases of the solutions of $F(r) = 0$: r_1, r_2: Real and distinct roots, Complex roots and Equal roots

5.3.1 Real and distinct roots

$$y(t) = c_1 t^{r_1} + c_2 t^{r_2}$$

5.3.2 Complex roots

Suppose

$$r_{1,2} = \lambda \pm i\mu, \qquad \mu \neq 0.$$

with

$$\lambda = \frac{1-\alpha}{2}, \quad \mu = \frac{[4\beta - (\alpha-1)^2]^{\frac{1}{2}}}{2}$$

Then

$$\begin{aligned} t^{\lambda + i\mu} &= t^\lambda t^{i\mu} \\ &= t^\lambda e^{i\mu \ln t} \\ &= t^\lambda \left[\cos(\mu \ln t) + i \sin(\mu \ln t)\right]. \end{aligned}$$

The real and imaginary parts give

$$y_1 = t^\lambda \cos(\mu \ln t), \qquad y_2 = t^\lambda \sin(\mu \ln t).$$

Thus,

$$\boxed{y(t) = t^\lambda \left[c_1 \cos(\mu \ln t) + c_2 \sin(\mu \ln t)\right]}.$$

5.3.3 Equal roots

Suppose

$$F(r) = (r - r_1)^2.$$

We obtain only one solution t^{r_1} directly. To find the second one, one natural method is reduction of the order, one could read 3.3 to recall this method.
However, it's a bit complex (but absolutly solvable). There is another method: start with

$$\begin{aligned}\frac{\partial}{\partial r} L[y] &= \frac{\partial}{\partial r} h(\frac{\partial^2}{\partial t^2} y, \frac{\partial}{\partial r} y, y)\\ &\text{(as } h \text{ is a linear operator of } y'', y', y)\\ &= h(\frac{\partial^2}{\partial t^2}\frac{\partial}{\partial r} y, \frac{\partial}{\partial t}\frac{\partial}{\partial r} y, y)\\ &= L[\frac{\partial}{\partial r} y]\end{aligned}$$

So

$$\frac{\partial}{\partial r} L[t^r] = L[t^r \ln t] \tag{37}$$

and

$$\frac{\partial}{\partial r} L[t^r] = \frac{\partial}{\partial r}\Big((r - r_0)^2 t^r\Big) = (r - r_1)^2 t^r \ln t + 2(r - r_1) t^r \tag{38}$$

Combining (37) and (38) gives:

$$L[t^r \ln t] = (r - r_1)^2 t^r \ln t + 2(r - r_1) t^r$$

Set $r = r_1$:

$$L[t^{r_1} \ln t] = 0$$

So

$$y_2 = t^{r_0} \ln t,$$

and

$$\boxed{y(t) = t^{r_0}(c_1 + c_2 \ln t)}.$$

5.3.4 The negative interval

All we've talked about is $t > 0$. Now we are curious about $t < 0$.
For $t < 0$, set

$$t = -x, \qquad x > 0, \qquad y(t) = u(x).$$

By the chain rule,

$$\frac{\mathrm{d}y}{\mathrm{d}t} = -\frac{\mathrm{d}u}{\mathrm{d}x}, \qquad \frac{\mathrm{d}^2y}{\mathrm{d}t^2} = \frac{\mathrm{d}^2u}{\mathrm{d}x^2}.$$

Therefore,

$$t^2\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + \alpha t\frac{\mathrm{d}y}{\mathrm{d}t} + \beta y = x^2\frac{\mathrm{d}^2u}{\mathrm{d}x^2} + \alpha x\frac{\mathrm{d}u}{\mathrm{d}x} + \beta u.$$

This is the same Euler equation in x. Since

$$x = -t = |t|,$$

the formulas for $t < 0$ are obtained by

$$t^r \longrightarrow |t|^r, \qquad \ln t \longrightarrow \ln|t|.$$

An Euler equation centered at t_0 has the form

$$(t-t_0)^2\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + \alpha(t-t_0)\frac{\mathrm{d}y}{\mathrm{d}t} + \beta y = 0.$$

With $x = t - t_0$, it reduces to the Euler equation centered at 0.

5.4 The Method of Frobenius

Recall the definition of regular singular points(2.5.2), consider

$$L[y] = \frac{\mathrm{d}^2y}{\mathrm{d}t^2} + p(t)\frac{\mathrm{d}y}{\mathrm{d}t} + q(t)y = 0. \tag{39}$$

where

$$p(t) = \frac{p_0}{t} + p_1 + p_2t + p_3t^2 + \cdots,$$
$$q(t) = \frac{q_0}{t^2} + \frac{q_1}{t} + q_2 + q_3t + \cdots.$$

Also consider

$$t^2\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + t\big(tp(t)\big)\frac{\mathrm{d}y}{\mathrm{d}t} + t^2q(t)y = 0. \tag{40}$$

So we could see *the Method of Frobenius*

$$t^2\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + t\big(p_0 + p_1t + p_2t^2 + \cdots\big)\frac{\mathrm{d}y}{\mathrm{d}t} + \big(q_0 + q_1t + q_2t^2 + \cdots\big)y = 0. \tag{41}$$

In order to fit the degree of $p(t)$ and $q(t)$, we could set:

$$y(t) = \sum_{n=0}^{\infty} a_nt^{n+r} = t^r\sum_{n=0}^{\infty} a_nt^n \tag{42}$$

Then

$$y' = \sum_{n=0}^{\infty}(n+r)a_n t^{n+r-1},$$
$$y'' = \sum_{n=0}^{\infty}(n+r)(n+r-1)a_n t^{n+r-2}.$$

Thus,

$$\begin{aligned}
L[y] = & \sum_{n=0}^{\infty}(n+r)(n+r-1)a_n t^{n+r} \\
& + \left(\sum_{m=0}^{\infty} p_m t^m\right)\left(\sum_{k=0}^{\infty}(k+r)a_k t^{k+r}\right) \\
& + \left(\sum_{m=0}^{\infty} q_m t^m\right)\left(\sum_{k=0}^{\infty} a_k t^{k+r}\right) \\
= t^r & \Bigg\{ \sum_{n=0}^{\infty}(n+r)(n+r-1)a_n t^n \\
& \quad + \left(\sum_{m=0}^{\infty} p_m t^m\right)\left(\sum_{n=0}^{\infty}(n+r)a_n t^n\right) \\
& \quad + \left(\sum_{m=0}^{\infty} q_m t^m\right)\left(\sum_{n=0}^{\infty} a_n t^n\right)\Bigg\}.
\end{aligned}$$

Thus,

$$\begin{aligned}
L[y] &= [r(r-1)+p_0 r+q_0]a_0 t^r \\
&\quad + \{[(1+r)r+p_0(1+r)+q_0]a_1+(rp_1+q_1)a_0\}\, t^{r+1} + \\
&\quad \vdots \\
&\quad + \left\{[(n+r)(n+r-1)+p_0(n+r)+q_0]a_n + \sum_{k=0}^{n-1}[(k+r)p_{n-k}+q_{n-k}]a_k\right\} t^{n+r} \\
&\quad + \cdots
\end{aligned}$$

Set

$$F(r) = r(r-1)+p_0 r+q_0.$$

Thus, we can get

$$\begin{aligned}
L[y] &= a_0 F(r) t^r \\
&\quad + \sum_{n=1}^{\infty}\left\{F(n+r)a_n + \sum_{k=0}^{n-1}\left[(k+r)p_{n-k}+q_{n-k}\right]a_k\right\} t^{n+r}.
\end{aligned}$$

The lowest power t^r gives

$$F(r) = r(r-1)+p_0 r+q_0 = 0 \tag{43}$$

(43) is called the *indicial* equation of (39).
For $n \geq 1$,

$$F(n+r)a_n = -\sum_{k=0}^{n-1} \left[(k+r)p_{n-k} + q_{n-k}\right] a_k, \tag{44}$$

So we just need to analysis (43). Assume it has roots r_1 and r_2. It maybe real and distinct roots, complex roots and equal roots. One more difficult thing is that even we can solve two real and distinct roots, it's still very bad if $r_2 - r_1$ is a integer.

5.4.1 Real and distinct roots without integer difference

Let r_1, r_2 be the roots of $F(r) = 0$, with $r_1 > r_2$ when they are real and $r_2 - r_1 \notin Z$. Since

$$F(r) = (r - r_1)(r - r_2),$$

so $F(r_i + n)$ will never be zero when $n \geq 1$.

By (44), we can get:

$$a_n = -\frac{\displaystyle\sum_{k=0}^{n-1} \left[(k+r)p_{n-k} + q_{n-k}\right] a_k}{F(n+r)}, \qquad n \geq 1 \tag{45}$$

And the solutions:

$$y_1(t) = t^{r_1} \sum_{n=0}^{\infty} a_n t^n, \qquad y_2(t) = t^{r_2} \sum_{n=0}^{\infty} b_n t^n. \tag{46}$$

5.4.2 Complex

If an indicial root is complex, the Frobenius series

$$y(t) = t^r \sum_{n=0}^{\infty} a_n t^n,$$

is complex-valued. Both its real and imaginary parts of $y(t)$ are real-valued solutions of (39) For real p and q,

$$\operatorname{Re} y \qquad \text{and} \qquad \operatorname{Im} y$$

are real-valued solutions.

5.4.3 Equal roots

Suppose

$$r_1 = r_2 = r_0.$$

The direct recurrence gives one solution

$$y_1(t) = t^{r_0} \sum_{n=0}^{\infty} a_n(r_0) t^n.$$

Keep r free and write

$$y(t,r) = t^r \sum_{n=0}^{\infty} a_n(r) t^n.$$

For $n \geq 1$, choose

$$a_n(r) = -\frac{\displaystyle\sum_{k=0}^{n-1} \left[(k+r)p_{n-k} + q_{n-k}\right] a_k(r)}{F(n+r)}.$$

This makes every coefficient of t^{n+r}, $n \geq 1$, equal to zero. Therefore,

$$L[y(t,r)] = a_0 F(r) t^r.$$

For an equal root,

$$F(r) = (r - r_0)^2.$$

Differentiate with respect to r:

$$\begin{aligned} L\left[\frac{\partial y}{\partial r}(t,r)\right] &= a_0 \frac{\partial}{\partial r}\left[F(r)t^r\right] \\ &= a_0 \left[F'(r)t^r + F(r)t^r \ln t\right]. \end{aligned}$$

At $r = r_0$,

$$F(r_0) = F'(r_0) = 0.$$

Hence,

$$y_2(t) = \left.\frac{\partial}{\partial r} y(t,r)\right|_{r=r_0}$$

is also a solution. Expanding the derivative,

$$\begin{aligned} y_2(t) &= \left.\frac{\partial}{\partial r}\left[t^r \sum_{n=0}^{\infty} a_n(r) t^n\right]\right|_{r=r_0} \\ &= t^{r_0} \ln t \sum_{n=0}^{\infty} a_n(r_0) t^n + t^{r_0} \sum_{n=0}^{\infty} a_n'(r_0) t^n \\ &= y_1(t) \ln t + t^{r_0} \sum_{n=0}^{\infty} a_n'(r_0) t^n. \end{aligned}$$

Thus,

$$\boxed{\begin{aligned} &y_1(t) = t^{r_0} \sum_{n=0}^{\infty} a_n t^n, \\ &y_2(t) = y_1(t) \ln t + t^{r_0} \sum_{n=0}^{\infty} b_n t^n. \end{aligned}}$$

5.4.4 Roots differing by a positive integer

Suppose

$$r_1 = r_2 + N, \qquad N \in \mathbb{N}.$$

The larger root gives

$$y_1(t) = t^{r_1} \sum_{n=0}^{\infty} a_n t^n.$$

For the smaller root, at $n = N$, set

$$S_N := \sum_{k=0}^{N-1} \left[(k + r_2)p_{N-k} + q_{N-k}\right] a_k,$$

$$F(N + r_2)a_N = -S_N, \qquad F(N + r_2) = F(r_1) = 0.$$

Hence the recurrence becomes $0 \cdot a_N = -S_N$. If $S_N = 0$, a pure Frobenius series may exist and the logarithmic coefficient may be zero. If $S_N \neq 0$, a logarithmic term is necessary.

The second solution has the form

$$\boxed{y_2(t) = C\, y_1(t) \ln t + t^{r_2} \sum_{n=0}^{\infty} b_n t^n},$$

where C may be zero.

In applications, only the first few coefficients b_n are usually needed.

5.4.5 negative interval

On an interval $t < 0$, we use

$$y(t) = |t|^r \sum_{n=0}^{\infty} a_n t^n.$$

Theorem 2.5.5. *Consider the differential equation (39) where $t = 0$ is a regular singular point. Then, the functions $tp(t)$ and $t^2q(t)$ are analytic at $t = 0$ with power series expansions*

$$tp(t) = p_0 + p_1 t + p_2 t^2 + \cdots, \qquad t^2 q(t) = q_0 + q_1 t + q_2 t^2 + \cdots$$

which converge for $|t| < \rho$. Let r_1 and r_2 be the two roots of the indicial equation

$$r(r-1) + p_0 r + q_0 = 0$$

with $r_1 \geq r_2$ if they are real. Then, Equation (39) has two linearly independent solutions $y_1(t)$ and $y_2(t)$ on the interval $0 < t < \rho$ of the following form:

(a) If $r_1 - r_2$ is not a positive integer, then

$$y_1(t) = t^{r_1} \sum_{n=0}^{\infty} a_n t^n, \qquad y_2(t) = t^{r_2} \sum_{n=0}^{\infty} b_n t^n.$$

(b) If $r_1 = r_2$, then

$$y_1(t) = t^{r_1} \sum_{n=0}^{\infty} a_n t^n, \qquad y_2(t) = y_1(t) \ln t + t^{r_1} \sum_{n=0}^{\infty} b_n t^n.$$

(c) If $r_1 - r_2 = N$, a positive integer, then

$$y_1(t) = t^{r_1} \sum_{n=0}^{\infty} a_n t^n, \qquad y_2(t) = a y_1(t) \ln t + t^{r_2} \sum_{n=0}^{\infty} b_n t^n$$

where the constant a may turn out to be zero.

6 The method of variantion of parameters

Consider

$$L[y](t) = \frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + p(t)\frac{\mathrm{d}y}{\mathrm{d}t} + q(t)y = g(t), \tag{47}$$

$$L[y](t) = \frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + p(t)\frac{\mathrm{d}y}{\mathrm{d}t} + q(t)y = 0 \tag{48}$$

Let y_1 and y_2 be two linear independent solutions of (48), and assume we've *known* them:

$$L[y_1] = 0, \quad L[y_2] = 0$$

Let ψ be a particular solution of (47).
We may write ψ in the form:

$$\psi(t) = u_1(t)y_1(t) + u_2(t) + y_2(t) \tag{49}$$

It may be weird, as the problem of finding one function becomes finding two functions. But it gives more freedom to us. We can perhaps give more assmuption.
Now we would calculate $L[\psi]$:

$$\psi' = (u_1y_1 + u_2y_2)' = u_1y_1' + u_2y_2' + u_1'y_1 + u_2'y_2.$$

$$\begin{aligned}\psi'' &= (u_1y_1')' + (u_2y_2')' + (u_1'y_1)' + (u_2'y_2)' \\ &= u_1y_1'' + u_2y_2'' + 2u_1y_1' + 2u_2y_2' + u_1'y_1 + u_2'y_2.\end{aligned}$$

$$\begin{aligned}L[\psi] &= \psi'' + p\psi' + q\psi \\ &= u_1y_1'' + u_2y_2'' + 2u_1y_1' + 2u_2y_2' + u_1'y_1 + u_2y_2 \\ &\quad + p(u_1y_1' + u_2y_2' + u_1'y_1 + u_2y_1) + q(u_1y_1 + u_2y_2) \\ &= u_1\left(y_1'' + py_1' + qy_1\right) + u_2\left(y_2'' + py_2' + qy_2\right) \\ &\quad + y_1u_1'' + y_2u_2'' + (2y_1' + py_1)u_1' + (2y_2' + py_2)u_2' \\ &= y_1u_1'' + y_2u_2'' + (2y_1' + py_1)u_1' + (2y_2' + py_2)u_2'\end{aligned}$$

Observe the *troublesome* terms:

$$y_1u_1'' + y_2u_2''.$$

By its form, we may recall:

$$\frac{d}{dt}(y_1u_1' + y_2u_2') = y_1'u_1' + y_1u_1'' + y_2'u_2' + y_2u_2''.$$

$$y_1u_1'' + y_2u_2'' = \frac{d}{dt}(y_1u_1' + y_2u_2') - y_1'u_1' - y_2'u_2'.$$

Substituting this back into $L[\psi]$:

$$
\begin{aligned}
L[\psi] &= \frac{d}{dt}(y_1u_1' + y_2u_2') - y_1'u_1' - y_2'u_2' + 2y_1'u_1' + 2y_2'u_2' + p(y_1u_1' + y_2u_2') \\
&= \frac{d}{dt}(y_1u_1' + y_2u_2') + p(t)(y_1u_1' + y_2u_2') + y_1'u_1' + y_2'u_2'.
\end{aligned}
$$

One natural assumption idea is to set $\boxed{y_1u_1' + y_2u_2' = 0}$, and:

$$L[\psi] = y_1'u_1' + y_2'u_2'.$$

So the problem is to find u_1 and u_2 s.t.

$$y_1u_1' + y_2u_2' = 0 \tag{50}$$

$$y_1'u_1' + y_2'u_2' = g(t) \tag{51}$$

By some very basic transform:

$$\boxed{u_1'(t) = -\frac{g(t)y_2(t)}{W[y_1, y_2](t)} \quad \text{and} \quad u_2'(t) = \frac{g(t)y_1(t)}{W[y_1, y_2](t)}} \tag{52}$$

where

$$W[y_1, y_2](t) = y_1(t)y_2'(t) - y_1'(t)y_2(t)$$

Then, by integrating the right-hand-side, we can get u_1 and u_2.

Chapter 3

Systems of Differential Equations

1 Algebraic properties of solutions of linear systems

1.1 Systems of first-order differential equations

Consider n unknown functions

$$x_1(t), x_2(t), \ldots, x_n(t).$$

A system of n first-order differential equations has the form

$$\begin{cases} \frac{\mathrm{d}x_1}{\mathrm{d}t} = f_1(t, x_1, \ldots, x_n), \\ \frac{\mathrm{d}x_2}{\mathrm{d}t} = f_2(t, x_1, \ldots, x_n), \\ \qquad \vdots \\ \frac{\mathrm{d}x_n}{\mathrm{d}t} = f_n(t, x_1, \ldots, x_n). \end{cases} \tag{1}$$

Definition 3.1.1 (Solution of a first-order system)**.**
A solution of (1) *is a collection of* n *functions*

$$x_1(t), \ldots, x_n(t)$$

such that

$$\frac{\mathrm{d}x_j}{\mathrm{d}t} = f_j\big(t, x_1(t), \ldots, x_n(t)\big), \qquad j = 1, \ldots, n.$$

We may also prescribe the values of all components at the same time t_0:

$$x_1(t_0) = x_1^0, \quad x_2(t_0) = x_2^0, \quad \ldots, \quad x_n(t_0) = x_n^0. \tag{2}$$

Equations (1) and (2) together form an initial-value problem.

1.2 Higher-order equations as first-order systems

A higher-order equation for one function can be rewritten as a first-order system. Consider

$$a_n(t)y^{(n)} + a_{n-1}(t)y^{(n-1)} + \cdots + a_1(t)y' + a_0(t)y = 0, \qquad a_n(t) \neq 0. \tag{3}$$

Set

$$x_1 = y, \qquad x_2 = y', \qquad \ldots, \qquad x_n = y^{(n-1)}.$$

Then

$$\frac{\mathrm{d}x_1}{\mathrm{d}t} = x_2, \qquad \frac{\mathrm{d}x_2}{\mathrm{d}t} = x_3, \qquad \ldots, \qquad \frac{\mathrm{d}x_{n-1}}{\mathrm{d}t} = x_n.$$

Equation (3) determines the final derivative:

$$\frac{\mathrm{d}x_n}{\mathrm{d}t} = -\frac{a_0(t)}{a_n(t)}x_1 - \frac{a_1(t)}{a_n(t)}x_2 - \cdots - \frac{a_{n-1}(t)}{a_n(t)}x_n.$$

Thus, (3) is equivalent to

$$\begin{cases} \frac{\mathrm{d}x_1}{\mathrm{d}t} = x_2, \\ \frac{\mathrm{d}x_2}{\mathrm{d}t} = x_3, \\ \qquad \vdots \\ \frac{\mathrm{d}x_{n-1}}{\mathrm{d}t} = x_n, \\ \dfrac{\mathrm{d}x_n}{\mathrm{d}t} = -\displaystyle\sum_{j=1}^{n} \frac{a_{j-1}(t)}{a_n(t)}x_j. \end{cases} \tag{4}$$

1.3 Linear systems

Definition 3.1.2 (Linear system of n first-order equations)**.**
A system is linear if it has the form

$$\begin{cases} \frac{\mathrm{d}x_1}{\mathrm{d}t} = a_{11}(t)x_1 + \cdots + a_{1n}(t)x_n + g_1(t), \\ \frac{\mathrm{d}x_2}{\mathrm{d}t} = a_{21}(t)x_1 + \cdots + a_{2n}(t)x_n + g_2(t), \\ \qquad \vdots \\ \frac{\mathrm{d}x_n}{\mathrm{d}t} = a_{n1}(t)x_1 + \cdots + a_{nn}(t)x_n + g_n(t). \end{cases} \tag{5}$$

Definition 3.1.3 (Homogeneous linear system)**.**
The linear system (5) *is homogeneous if*

$$g_1(t) = g_2(t) = \cdots = g_n(t) = 0.$$

Definition 3.1.4 (Nonhomogeneous linear system)**.**
The linear system (5) *is nonhomogeneous if at least one g_j is not identically zero.*

We first study homogeneous systems with constant coefficients:

$$\begin{cases} \frac{\mathrm{d}x_1}{\mathrm{d}t} = a_{11}x_1 + \cdots + a_{1n}x_n, \\ \frac{\mathrm{d}x_2}{\mathrm{d}t} = a_{21}x_1 + \cdots + a_{2n}x_n, \\ \qquad \vdots \\ \frac{\mathrm{d}x_n}{\mathrm{d}t} = a_{n1}x_1 + \cdots + a_{nn}x_n. \end{cases} \tag{6}$$

Writing all n equations separately becomes inconvenient. We now introduce vectors and matrices to put (6) into one equation.

1.4 Vectors, matrices and the matrix form of a linear system

Definition 3.1.5 (Vector)**.**
A vector in $\mathbb{R}^n$ is an ordered list of n numbers:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

The numbers $x_1, \ldots, x_n$ are the components of $\mathbf{x}$.

Definition 3.1.6 (Vector-valued function)**.**
If every component depends on t, then

$$\mathbf{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix}$$

is a vector-valued function. Its derivative is defined componentwise:

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = \begin{pmatrix} \frac{\mathrm{d}x_1}{\mathrm{d}t} \\ \frac{\mathrm{d}x_2}{\mathrm{d}t} \\ \vdots \\ \frac{\mathrm{d}x_n}{\mathrm{d}t} \end{pmatrix}. \tag{7}$$

Definition 3.1.7 (Matrix)**.**
An $m \times n$ matrix is an array

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

The entry a_{ij} lies in row i and column j. If $m = n$, then A is a square matrix.

Definition 3.1.8 (Matrix-vector product)**.**
Let $A = (a_{ij})$ be an $n \times n$ matrix and $\mathbf{x} = (x_1, \dots, x_n)^{\mathrm{T}}$. The i-th component of $A\mathbf{x}$ is

$$(A\mathbf{x})_i = \sum_{j=1}^{n} a_{ij} x_j.$$

Equivalently,

$$A\mathbf{x} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} \sum_{j=1}^{n} a_{1j} x_j \\ \sum_{j=1}^{n} a_{2j} x_j \\ \vdots \\ \sum_{j=1}^{n} a_{nj} x_j \end{pmatrix}. \tag{8}$$

Let

$$\mathbf{x}(t) = \begin{pmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{pmatrix}, \qquad A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix}.$$

By the definition of matrix multiplication, the right-hand side of (6) is exactly $A\mathbf{x}$. Therefore, the system becomes

$$\boxed{\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = A\mathbf{x}} \tag{9}$$

Similarly, the initial conditions $x_1(t_0) = x_1^0, \quad \dots, \quad x_n(t_0) = x_n^0$ can be written as

$$\mathbf{x}(t_0) = \mathbf{x}^0, \qquad \mathbf{x}^0 = \begin{pmatrix} x_1^0 \\ \vdots \\ x_n^0 \end{pmatrix}.$$

Hence, the initial-value problem is

$$\boxed{\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = A\mathbf{x}, \qquad \mathbf{x}(t_0) = \mathbf{x}^0} \tag{10}$$

Lemma 1.1. *Let A be an $n \times n$ matrix, let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, and let $c \in \mathbb{R}$. Then*

$$A(c\mathbf{x}) = cA\mathbf{x}$$

and

$$A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y}.$$

Theorem 3.1.2. *Let $\mathbf{x}(t)$ and $\mathbf{y}(t)$ be solutions of*

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = A\mathbf{x}.$$

Then:

1. $c\mathbf{x}(t)$ *is a solution for every constant* c*;*
2. $\mathbf{x}(t) + \mathbf{y}(t)$ *is also a solution.*

2 Linear algebra to differential equations

Consider

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = A\mathbf{x}, \quad \mathbf{x}(t) = \begin{pmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{pmatrix}, \qquad A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix}. \tag{11}$$

Theorem 3.2.1 (Existence–uniqueness theorem)**.** *There exists one, and only one, solution of the initial-value problem*

$$\frac{dx}{dt} = Ax, \qquad x(t_0) = x^0 = \begin{bmatrix} x_1^0 \\ x_2^0 \\ \vdots \\ x_n^0 \end{bmatrix}. \tag{12}$$

Moreover, this solution exists for $-\infty < t < \infty$.

Theorem 4 is an extremely powerful theorem, and has many implications. In particular, if $x(t)$ is a nontrivial solution, then $x(t) \neq 0$ for any t. (If $x(t^*) = 0$ for some t^*, then $x(t)$ must be identically zero, since it, and the trivial solution, satisfy the same differential equation and have the same value at $t = t^*$.)
The *proof* of Theorem 3.2.1 will be shown later.

Theorem 3.2.2. *Let $I \subseteq \mathbb{R}$ be an interval, let*

$$A : I \to \mathbb{R}^{n\times n}$$

be continuous, and fix $t_0 \in I$. Consider (11). Let V denote the vector space of all solutions $x : I \to \mathbb{R}^n$ of this system. Then

$$\dim V = n.$$

Theorem 3.2.3. *Let $\psi^{(1)}, \ldots, \psi^{(m)}$ be solutions of*

$$x'(t) = A(t)x(t)$$

on an interval I, where A is continuous. The following statements are equivalent:

1. *The functions $\psi^{(1)}, \ldots, \psi^{(m)}$ are linearly independent in the solution space.*
2. *There exists $t_0 \in I$ such that*

$$\psi^{(1)}(t_0), \ldots, \psi^{(m)}(t_0)$$

are linearly independent in $\mathbb{R}^n$.

3. For every $t \in I$, the vectors

$$\psi^{(1)}(t), \ldots, \psi^{(m)}(t)$$

are linearly independent in $\mathbb{R}^n$.

2.1 proof of Theorem 3.2.2

Theorem 3.2.2. *Let $I \subseteq \mathbb{R}$ be an interval, let*

$$A : I \to \mathbb{R}^{n \times n}$$

be continuous, and fix $t_0 \in I$. Consider (11). Let V denote the vector space of all solutions $x : I \to \mathbb{R}^n$ of this system. Then

$$\dim V = n.$$

Proof. **MAP**

$$\text{Prove } \boxed{\phi^1, \cdots, \phi^n \text{ linear independent}}$$

$$\Updownarrow$$

$$\text{Prove } \boxed{\text{If } c_1, \cdots, c_n \text{ satisfies } c_1\phi^1 + \cdots + c_n\phi^n = 0, \text{ then } c_1 = \cdots = c_n = 0.}$$

$$\Updownarrow$$

$$\boxed{\text{As } c_1\phi^1 + \cdots + c_n\phi^n = 0 \text{ holds for all } t, \text{ it must include } t = 0.}$$

$$\Updownarrow$$

$$\boxed{\text{We construct } n \text{ linear independent } \phi_0^i}$$

Let $e_1, \ldots, e_n$ be the standard basis of $\mathbb{R}^n$. For each $j = 1, \ldots, n$, let

$$\phi^{(j)} : I \to \mathbb{R}^n$$

be the unique solution of the initial-value problem

$$\begin{cases} \left(\phi^{(j)}\right)'(t) = A(t)\phi^{(j)}(t), \\ \phi^{(j)}(t_0) = e_j. \end{cases}$$

We first prove that

$$\phi^{(1)}, \ldots, \phi^{(n)}$$

are linearly independent in V. Suppose that

$$\sum_{j=1}^{n} c_j \phi^{(j)} = 0_V$$

for some constants $c_1, \ldots, c_n \in \mathbb{R}$, where 0_V denotes the zero solution. By equality of functions, this means that

$$\sum_{j=1}^{n} c_j \phi^{(j)}(t) = 0 \qquad \text{for every } t \in I.$$

In particular, evaluating at $t = t_0$, we obtain

$$0 = \sum_{j=1}^{n} c_j \phi^{(j)}(t_0) = \sum_{j=1}^{n} c_j e_j.$$

Since $e_1, \ldots, e_n$ are linearly independent in $\mathbb{R}^n$, it follows that

$$c_1 = \cdots = c_n = 0.$$

Therefore,

$$\phi^{(1)}, \ldots, \phi^{(n)}$$

are linearly independent in V.

We next prove that they span V. Let $x \in V$ be arbitrary. Since $e_1, \ldots, e_n$ form a basis of $\mathbb{R}^n$, there exist unique constants $c_1, \ldots, c_n$ such that

$$x(t_0) = \sum_{j=1}^{n} c_j e_j.$$

Define

$$y(t) = \sum_{j=1}^{n} c_j \phi^{(j)}(t).$$

By linearity of the differential equation,

$$\begin{aligned} y'(t) &= \sum_{j=1}^{n} c_j \left(\phi^{(j)}\right)'(t) \\ &= \sum_{j=1}^{n} c_j A(t) \phi^{(j)}(t) \\ &= A(t) \sum_{j=1}^{n} c_j \phi^{(j)}(t) \\ &= A(t) y(t). \end{aligned}$$

Moreover,

$$y(t_0) = \sum_{j=1}^{n} c_j \phi^{(j)}(t_0) = \sum_{j=1}^{n} c_j e_j = x(t_0).$$

Thus x and y solve the same initial-value problem. By uniqueness of solutions,

$$x(t) = y(t) = \sum_{j=1}^{n} c_j \phi^{(j)}(t) \qquad \text{for every } t \in I.$$

Hence
$$V = \operatorname{span}\left\{\phi^{(1)}, \ldots, \phi^{(n)}\right\}.$$
Therefore,
$$\left\{\phi^{(1)}, \ldots, \phi^{(n)}\right\}$$
is a basis of V, and consequently
$$\dim V = n.$$
□

2.2 proof of Theorem 3.2.3

Theorem 3.2.3. *Let $\psi^{(1)}, \ldots, \psi^{(m)}$ be solutions of*
$$x'(t) = A(t)x(t)$$
on an interval I, where A is continuous. The following statements are equivalent:

1. *The functions $\psi^{(1)}, \ldots, \psi^{(m)}$ are linearly independent in the solution space.*
2. *There exists $t_0 \in I$ such that*
$$\psi^{(1)}(t_0), \ldots, \psi^{(m)}(t_0)$$
are linearly independent in $\mathbb{R}^n$.
3. *For every $t \in I$, the vectors*
$$\psi^{(1)}(t), \ldots, \psi^{(m)}(t)$$
are linearly independent in $\mathbb{R}^n$.

Proof. We first prove that $2 \Rightarrow 1$. Suppose that, for some constants $c_1, \ldots, c_m$,
$$\sum_{j=1}^{m} c_j \psi^{(j)} = 0$$
as an equality of functions. Evaluating at $t = t_0$, we obtain
$$\sum_{j=1}^{m} c_j \psi^{(j)}(t_0) = 0.$$
Since the vectors $\psi^{(1)}(t_0), \ldots, \psi^{(m)}(t_0)$ are linearly independent, we have
$$c_1 = \cdots = c_m = 0.$$
Hence the solution functions are linearly independent.

We next prove that $1 \Rightarrow 3$. Fix any $t_1 \in I$ and suppose that

$$\sum_{j=1}^{m} c_j \psi^{(j)}(t_1) = 0.$$

Define

$$z(t) = \sum_{j=1}^{m} c_j \psi^{(j)}(t).$$

By linearity,

$$z'(t) = A(t)z(t),$$

and by assumption,

$$z(t_1) = 0.$$

The zero function is also a solution of

$$z'(t) = A(t)z(t), \qquad z(t_1) = 0.$$

By uniqueness of the initial-value problem,

$$z(t) \equiv 0 \qquad \text{on } I.$$

Since the functions $\psi^{(1)}, \ldots, \psi^{(m)}$ are linearly independent, it follows that

$$c_1 = \cdots = c_m = 0.$$

Thus the vectors $\psi^{(1)}(t_1), \ldots, \psi^{(m)}(t_1)$ are linearly independent. Since t_1 was arbitrary, they are linearly independent for every $t \in I$.
Finally, $3 \Rightarrow 2$ is immediate. $\square$

3 The eigenvalue-eigenvector method

Consider

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = A\mathbf{x}, \qquad \mathbf{x}(t) = \begin{pmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{pmatrix}, \qquad A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix}. \tag{13}$$

Our goal is to find n linearly independent solutions.
One natural idea is to try

$$\mathbf{x}(t) = e^{\lambda t}\mathbf{v}, \qquad \mathbf{v} \neq \mathbf{0}, \tag{14}$$

where $\mathbf{v}$ is a constant vector.
Then

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = \lambda e^{\lambda t}\mathbf{v} \quad A\mathbf{x} = A\left(e^{\lambda t}\mathbf{v}\right) = e^{\lambda t}A\mathbf{v}.$$

Plugging (14) into (13) gives

$$\lambda e^{\lambda t}\mathbf{v} = e^{\lambda t}A\mathbf{v}$$
$$\Downarrow$$
$$A\mathbf{v} = \lambda\mathbf{v}.$$

Therefore, we only need to solve the eigenvalue and eigenvector of A.

Definition 3.3.1 (Characteristic polynomial)**.**
The polynomial

$$p_A(\lambda) = \det(A - \lambda I) \tag{15}$$

is called the characteristic polynomial of A. The equation

$$p_A(\lambda) = 0$$

is called the characteristic equation.

Suppose that A has n linearly independent eigenvectors

$$\mathbf{v}_1, \dots, \mathbf{v}_n$$

with corresponding eigenvalues *(need not to be distinct)*.

$$\lambda_1, \dots, \lambda_n.$$

And we may write:

$$\mathbf{x}_j(t) = e^{\lambda_j t}\mathbf{v}_j, \qquad j = 1, \dots, n,$$

are solutions. Moreover,

$$\mathbf{x}_j(0) = \mathbf{v}_j.$$

Theorem 3.2.3 implies that $\mathbf{x}_1, \dots, \mathbf{x}_n$ are linearly independent. Since the solution space has dimension n,

$$\boxed{\mathbf{x}(t) = c_1 e^{\lambda_1 t}\mathbf{v}_1 + \cdots + c_n e^{\lambda_n t}\mathbf{v}_n.} \tag{16}$$

3.1 Real and distinct roots

The simplest case is

$$\lambda_1, \ldots, \lambda_n \in \mathbb{R}, \qquad \lambda_i \neq \lambda_j \quad (i \neq j).$$

We first need to know that the corresponding eigenvectors are linearly independent.

Theorem 3.3.1. *Eigenvectors corresponding to distinct eigenvalues are linearly independent.*

Proof. We use induction on the number of eigenvectors. The result is immediate for one eigenvector.
Suppose it is true for $k-1$ eigenvectors. Let $\mathbf{v}_1, \ldots, \mathbf{v}_k$ correspond to the distinct eigenvalues $\lambda_1, \ldots, \lambda_k$, and suppose

$$c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = \mathbf{0}. \tag{17}$$

Applying A gives

$$c_1\lambda_1\mathbf{v}_1 + \cdots + c_k\lambda_k\mathbf{v}_k = \mathbf{0}.$$

Subtracting λ_1 times (17), we obtain

$$c_2(\lambda_2 - \lambda_1)\mathbf{v}_2 + \cdots + c_k(\lambda_k - \lambda_1)\mathbf{v}_k = \mathbf{0}.$$

By the induction hypothesis,

$$c_j(\lambda_j - \lambda_1) = 0, \qquad j = 2, \ldots, k.$$

Since the eigenvalues are distinct,

$$c_2 = \cdots = c_k = 0.$$

Equation (17) then gives

$$c_1 = 0.$$

Thus, $\mathbf{v}_1, \ldots, \mathbf{v}_k$ are linearly independent. □

Therefore, if A has n real and distinct eigenvalues, then

$$\boxed{\mathbf{x}(t) = \sum_{j=1}^{n} c_j e^{\lambda_j t}\mathbf{v}_j.} \tag{18}$$

3.2 Complex roots

Suppose A is real and the eigenvalue:

$$\lambda = \alpha + i\beta, \qquad \beta \neq 0,$$

the eigenvector:

$$\mathbf{v} = \mathbf{p} + i\mathbf{q}, \qquad \mathbf{p}, \mathbf{q} \in \mathbb{R}^n.$$

Then

$$\mathbf{x}(t) = e^{(\alpha+i\beta)t}(\mathbf{p} + i\mathbf{q})$$

is the solution. But we need to get the real solution.

Lemma 3.2. *If*

$$\mathbf{x}(t) = \mathbf{y}(t) + i\mathbf{z}(t)$$

is a complex-valued solution of $\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = A\mathbf{x}$, *where* A *is real, then both* $\mathbf{y}(t)$ *and* $\mathbf{z}(t)$ *are real-valued solutions.*

$$\begin{aligned} e^{(\alpha+i\beta)t}(\mathbf{p} + i\mathbf{q}) &= e^{\alpha t}\big(\cos(\beta t) + i\sin(\beta t)\big)(\mathbf{p} + i\mathbf{q}) \\ &= e^{\alpha t}\Big[\mathbf{p}\cos(\beta t) - \mathbf{q}\sin(\beta t) \\ &\qquad\qquad + i\big(\mathbf{p}\sin(\beta t) + \mathbf{q}\cos(\beta t)\big)\Big]. \end{aligned}$$

Therefore,

$$\boxed{\begin{aligned} \mathbf{y}(t) &= e^{\alpha t}\big(\mathbf{p}\cos(\beta t) - \mathbf{q}\sin(\beta t)\big), \\ \mathbf{z}(t) &= e^{\alpha t}\big(\mathbf{p}\sin(\beta t) + \mathbf{q}\cos(\beta t)\big) \end{aligned}} \tag{19}$$

are real-valued solutions.
The conjugate vector also gives the solution.

3.3 Equal roots

If the characteristic polynomial does not have n distinct roots, the matrix may fail to have n linearly independent eigenvectors. If it still has n linearly independent eigenvectors, then (16) already gives the general solution.
The remaining problem is:

$$\boxed{A \text{ has fewer than } n \text{ linearly independent eigenvectors.}}$$

Then solutions of the form $e^{\lambda t}\mathbf{v}$ are not enough.
For the scalar equation

$$\frac{\mathrm{d}x}{\mathrm{d}t} = ax,$$

we have

$$x(t) = e^{at}c.$$

This suggests that the system should have the form

$$\mathbf{x}(t) = e^{At}\mathbf{v}.$$

We first define e^{At}.

Definition 3.3.2 (Matrix exponential)**.**
For an $n \times n$ matrix A,

$$e^{At} := I + At + \frac{A^2t^2}{2!} + \cdots + \frac{A^nt^n}{n!} + \cdots. \tag{20}$$

The series converges for every t and can be differentiated term by term:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}e^{At} &= A + A^2t + \frac{A^3t^2}{2!} + \cdots \\ &= A\left(I + At + \frac{A^2t^2}{2!} + \cdots\right) \\ &= Ae^{At}.\end{aligned}$$

Thus,

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(e^{At}\mathbf{v}\right) = Ae^{At}\mathbf{v} = A\left(e^{At}\mathbf{v}\right), \tag{21}$$

so $e^{At}\mathbf{v}$ is a solution for every constant vector $\mathbf{v}$.
Fix an eigenvalue λ, and set

$$B = A - \lambda I.$$

Since

$$A = \lambda I + B$$

and λI commutes with B,

$$e^{At} = e^{(\lambda I + B)t} = e^{\lambda t}e^{Bt} = e^{\lambda t}e^{(A-\lambda I)t}. \tag{22}$$

The crucial question is: when does the series

$$e^{(A-\lambda I)t}\mathbf{v}$$

become a finite sum?
Suppose

$$(A - \lambda I)^m\mathbf{v} = \mathbf{0}.$$

Then, for every $j \geq m$,

$$(A - \lambda I)^j\mathbf{v} = \mathbf{0}.$$

Therefore,

$$\boxed{e^{At}\mathbf{v} = e^{\lambda t}\left[\mathbf{v} + t(A-\lambda I)\mathbf{v} + \cdots + \frac{t^{m-1}}{(m-1)!}(A-\lambda I)^{m-1}\mathbf{v}\right].} \tag{23}$$

Definition 3.3.3 (Generalized eigenvector). *A nonzero vector* $\mathbf{v}$ *satisfying*

$$(A - \lambda I)^m \mathbf{v} = \mathbf{0}$$

for some $m \geq 1$ *is called a generalized eigenvector corresponding to* λ*. If* $m = 1$*, it is an ordinary eigenvector.*

Thus, the method is:

$$\boxed{\begin{array}{l}\text{solve } (A - \lambda I)\mathbf{v} = \mathbf{0}, \\ \text{then solve } (A - \lambda I)^2\mathbf{v} = \mathbf{0}, \\ \text{then solve } (A - \lambda I)^3\mathbf{v} = \mathbf{0}, \ \ldots\end{array}} \tag{24}$$

At each step, every new independent vector gives a new solution through (23).

Lemma 3.3. *Regard* A *as a matrix over* $\mathbb{C}$*, and let* λ *be a root of* p_A *with algebraic multiplicity* r*. Then, for some* $m \leq r$*,*

$$\dim \ker(A - \lambda I)^m = r.$$

This linear-algebra result guarantees that the process stops and produces enough independent solutions.

Theorem 3.3.4 (Cayley–Hamilton theorem). *Let*

$$p_A(\lambda) = p_0 + p_1\lambda + \cdots + p_n\lambda^n$$

be the characteristic polynomial of A*. Then*

$$\boxed{p_A(A) = p_0 I + p_1 A + \cdots + p_n A^n = 0.} \tag{25}$$

Proof. We cannot directly write

$$p_A(A) = \det(A - AI),$$

because the variable λ in $\det(A - \lambda I)$ is a scalar.
Let $C(\lambda)$ be the adjugate of $A - \lambda I$. Then

$$(A - \lambda I)C(\lambda) = p_A(\lambda)I.$$

Every entry of $C(\lambda)$ is a polynomial of degree at most $n - 1$, so

$$C(\lambda) = C_0 + C_1\lambda + \cdots + C_{n-1}\lambda^{n-1}.$$

Also,

$$C(\lambda)(A - \lambda I) = p_A(\lambda)I,$$

so every coefficient C_j commutes with A. Hence,

$$(A - \lambda I)\left(C_0 + C_1\lambda + \cdots + C_{n-1}\lambda^{n-1}\right) = p_A(\lambda)I.$$

This is an identity between matrix polynomials in the scalar variable λ. Comparing coefficients first, and only then replacing each λ^j by A^j, gives

$$\begin{aligned} p_A(A) &= (A - A)\left(C_0 + C_1A + \cdots + C_{n-1}A^{n-1}\right) \\ &= 0. \end{aligned}$$

□

4 Fundamental matrix solutions

Consider

$$\frac{d\mathbf{x}}{dt} = A\mathbf{x}. \tag{26}$$

Suppose

$$\mathbf{x}_1(t), \ldots, \mathbf{x}_n(t)$$

are n linearly independent solutions. Then every solution has the form

$$\mathbf{x}(t) = c_1\mathbf{x}_1(t) + \cdots + c_n\mathbf{x}_n(t).$$

Put these solutions into the columns of a matrix:

$$X(t) = \begin{pmatrix} \mathbf{x}_1(t) & \cdots & \mathbf{x}_n(t) \end{pmatrix}, \qquad \mathbf{c} = \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}.$$

Thus,

$$\boxed{\mathbf{x}(t) = X(t)\mathbf{c}.} \tag{27}$$

Definition 3.4.1 (Fundamental matrix solution)**.**
A matrix $X(t)$ is a fundamental matrix solution of (26) *if its columns are n linearly independent solutions of* (26)*.*

For a matrix-valued function

$$X(t) = \begin{pmatrix} x_{11}(t) & \cdots & x_{1n}(t) \\ \vdots & \ddots & \vdots \\ x_{n1}(t) & \cdots & x_{nn}(t) \end{pmatrix},$$

we differentiate every entry:

$$X'(t) = \begin{pmatrix} x'_{11}(t) & \cdots & x'_{1n}(t) \\ \vdots & \ddots & \vdots \\ x'_{n1}(t) & \cdots & x'_{nn}(t) \end{pmatrix}.$$

Lemma 4.1. *Fix t_0. A matrix $X(t)$ is a fundamental matrix solution of* (26) *if and only if*

$$X'(t) = AX(t) \qquad \textit{and} \qquad \det X(t_0) \neq 0. \tag{28}$$

Lemma 4.2. *The matrix exponential e^{At} is a fundamental matrix solution of* (26)*.*

Lemma 4.3. *Let $X(t)$ and $Y(t)$ be two fundamental matrix solutions of* (26)*.*

Then there exists a constant invertible matrix C such that

$$Y(t) = X(t)C. \tag{29}$$

We now connect an arbitrary fundamental matrix solution with the matrix exponential.

Theorem 3.4.4. *Let $X(t)$ be a fundamental matrix solution of* (26)*. Then*

$$\boxed{e^{At} = X(t)X^{-1}(0).} \tag{30}$$

More generally,

$$\boxed{e^{A(t-t_0)} = X(t)X^{-1}(t_0).} \tag{31}$$

Therefore, the solution of

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = A\mathbf{x}, \qquad \mathbf{x}(t_0) = \mathbf{x}^0,$$

can be written in either form

$$\mathbf{x}(t) = X(t)X^{-1}(t_0)\mathbf{x}^0$$

or

$$\boxed{\mathbf{x}(t) = e^{A(t-t_0)}\mathbf{x}^0.}$$

4.1 Proof of Lemma 4.1, 4.2, 4.3 and Theorem 3.4.4

Proof of Lemma 4.1

Proof. Write

$$X(t) = \begin{pmatrix}\mathbf{x}_1(t) & \cdots & \mathbf{x}_n(t)\end{pmatrix}.$$

Then

$$X'(t) = \begin{pmatrix}\mathbf{x}_1'(t) & \cdots & \mathbf{x}_n'(t)\end{pmatrix}$$

and

$$AX(t) = \begin{pmatrix}A\mathbf{x}_1(t) & \cdots & A\mathbf{x}_n(t)\end{pmatrix}.$$

Hence,

$$X'(t) = AX(t) \quad \Longleftrightarrow \quad \mathbf{x}_j'(t) = A\mathbf{x}_j(t), \quad j = 1, \ldots, n.$$

Also,

$$\det X(t_0) \neq 0$$

if and only if

$$\mathbf{x}_1(t_0), \ldots, \mathbf{x}_n(t_0)$$

are linearly independent. By Theorem 3.2.3, this is equivalent to the solution functions

$$\mathbf{x}_1, \ldots, \mathbf{x}_n$$

being linearly independent. □

Proof of Lemma 4.2

Proof. We have
$$\frac{\mathrm{d}}{\mathrm{d}t}e^{At} = Ae^{At}.$$
Moreover,
$$e^{A0} = I, \qquad \det(e^{A0}) = \det I = 1 \neq 0.$$
The result follows from Lemma 4.1. □

Proof of Lemma 4.3

Proof. Write
$$X(t) = \begin{pmatrix}\mathbf{x}_1(t) & \cdots & \mathbf{x}_n(t)\end{pmatrix}, \qquad Y(t) = \begin{pmatrix}\mathbf{y}_1(t) & \cdots & \mathbf{y}_n(t)\end{pmatrix}.$$
The columns of X form a basis of the solution space. Thus, for each j,
$$\mathbf{y}_j(t) = c_{1j}\mathbf{x}_1(t) + \cdots + c_{nj}\mathbf{x}_n(t),$$
where $c_{1j}, \ldots, c_{nj}$ are constants. Let
$$C = (c_{ij}).$$
The n column equations are exactly
$$Y(t) = X(t)C.$$
Since $Y(t_0)$ and $X(t_0)$ are invertible,
$$C = X^{-1}(t_0)Y(t_0)$$
is also invertible. □

Proof of Theorem 3.4.4

Proof. By Lemmas 4.2 and 4.3, there exists a constant matrix C such that
$$e^{At} = X(t)C.$$
At $t = 0$,
$$I = X(0)C, \qquad C = X^{-1}(0).$$
This gives (30).
For the general formula, apply (30) to the shifted matrix function
$$Y(s) = X(t_0 + s).$$
Then
$$e^{As} = X(t_0 + s)X^{-1}(t_0).$$
Setting $s = t - t_0$ gives (31). □

5 The nonhomogeneous equation; variation of parameters

Consider the nonhomogeneous initial-value problem

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = A\mathbf{x} + \mathbf{f}(t), \qquad \mathbf{x}(t_0) = \mathbf{x}^0. \tag{32}$$

Let $X(t)$ be a fundamental matrix solution of the corresponding homogeneous equation

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = A\mathbf{x}.$$

For the homogeneous equation, the constant vector $\mathbf{c}$ gives

$$\mathbf{x}(t) = X(t)\mathbf{c}.$$

For the nonhomogeneous equation, one natural idea is to let the parameters depend on t:

$$\mathbf{x}(t) = X(t)\mathbf{u}(t), \qquad \mathbf{u}(t) = \begin{pmatrix} u_1(t) \\ \vdots \\ u_n(t) \end{pmatrix}. \tag{33}$$

Then

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = X'(t)\mathbf{u}(t) + X(t)\mathbf{u}'(t).$$

Plugging (33) into (32), we obtain

$$X'(t)\mathbf{u}(t) + X(t)\mathbf{u}'(t) = AX(t)\mathbf{u}(t) + \mathbf{f}(t).$$

Since $X'(t) = AX(t)$,

$$X(t)\mathbf{u}'(t) = \mathbf{f}(t).$$

The matrix $X(t)$ is invertible for every t, so

$$\boxed{\mathbf{u}'(t) = X^{-1}(t)\mathbf{f}(t).} \tag{34}$$

Integrating from t_0 to t gives

$$\mathbf{u}(t) = \mathbf{u}(t_0) + \int_{t_0}^{t} X^{-1}(s)\mathbf{f}(s)\,\mathrm{d}s.$$

The initial condition gives

$$\mathbf{x}^0 = X(t_0)\mathbf{u}(t_0), \qquad \mathbf{u}(t_0) = X^{-1}(t_0)\mathbf{x}^0.$$

Therefore,

$$\boxed{\mathbf{x}(t) = X(t)X^{-1}(t_0)\mathbf{x}^0 + X(t)\int_{t_0}^{t} X^{-1}(s)\mathbf{f}(s)\,\mathrm{d}s.} \tag{35}$$

If we choose

$$X(t) = e^{At},$$

then

$$X^{-1}(s) = e^{-As}$$

and

$$e^{At}e^{-At_0} = e^{A(t-t_0)}.$$

Also,

$$e^{At}e^{-As} = e^{A(t-s)}.$$

Thus, (35) becomes

$$\boxed{\mathbf{x}(t) = e^{A(t-t_0)}\mathbf{x}^0 + \int_{t_0}^{t} e^{A(t-s)}\mathbf{f}(s)\,\mathrm{d}s.} \tag{36}$$

We can verify the integral term directly. Set

$$\boldsymbol{\psi}(t) = \int_{t_0}^{t} e^{A(t-s)}\mathbf{f}(s)\,\mathrm{d}s.$$

Then

$$\begin{aligned}\boldsymbol{\psi}'(t) &= \mathbf{f}(t) + \int_{t_0}^{t} Ae^{A(t-s)}\mathbf{f}(s)\,\mathrm{d}s \\ &= \mathbf{f}(t) + A\boldsymbol{\psi}(t).\end{aligned}$$

Also,

$$\boldsymbol{\psi}(t_0) = \mathbf{0}.$$

Hence, (36) satisfies both the differential equation and the initial condition.
The function

$$\boldsymbol{\psi}(t) = X(t)\int_{t_0}^{t} X^{-1}(s)\mathbf{f}(s)\,\mathrm{d}s \tag{37}$$

is one particular solution. Therefore, every solution of

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = A\mathbf{x} + \mathbf{f}(t)$$

has the form

$$\boxed{\mathbf{x}(t) = X(t)\mathbf{c} + \boldsymbol{\psi}(t).} \tag{38}$$

5.1 Exponential forcing

Consider

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = A\mathbf{x} + \mathbf{v}e^{ct}, \tag{39}$$

where $\mathbf{v}$ is a constant vector. We first try a particular solution of the same type:

$$\boldsymbol{\psi}(t) = \mathbf{b}e^{ct}.$$

Then

$$c\mathbf{b}e^{ct} = A\mathbf{b}e^{ct} + \mathbf{v}e^{ct},$$

so

$$\boxed{(cI - A)\mathbf{b} = \mathbf{v}.} \tag{40}$$

If c is not an eigenvalue of A, then $cI - A$ is invertible and

$$\mathbf{b} = (cI - A)^{-1}\mathbf{v}.$$

Thus,

$$\boldsymbol{\psi}(t) = (cI - A)^{-1}\mathbf{v}e^{ct}.$$

If c is an eigenvalue of A, the matrix $cI - A$ is singular, and (40) may have no solution. Then we enlarge the guess:

$$\boxed{\boldsymbol{\psi}(t) = e^{ct}\left(\mathbf{b}_0 + t\mathbf{b}_1 + \cdots + t^{k-1}\mathbf{b}_{k-1}\right)} \tag{41}$$

for a suitable k.

Chapter 4

Qualitative theory of differential equations

1 Stability

Consider a differential equation:

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = f(\mathbf{x}) \tag{1}$$

Also consider different starting points:

$$\mathbf{x}_0 \longrightarrow \phi(t), \qquad \phi(t_0) = \mathbf{x}_0,$$

$$\mathbf{y}_0 \longrightarrow \psi(t), \qquad \psi(t_0) = \mathbf{y}_0.$$

One natural question is: if $\mathbf{x}_0$ is close enough to $\mathbf{y}_0$, will $\phi(t)$ remain close to $\psi(t)$ as t increases? Thus, we come to the definition of *stability*.

Definition 4.1.1 (Stability).
The solution $\mathbf{x} = \phi(t)$ of (1) is stable if, for every $\varepsilon > 0$, there exists $\delta = \delta(\varepsilon) > 0$ such that

$$|\psi_j(t_0) - \phi_j(t_0)| < \delta, \quad j = 1, \ldots, n \quad \Big\} \Longrightarrow \begin{array}{r} |\psi_j(t) - \phi_j(t)| < \varepsilon, \\ j = 1, \ldots, n, \quad t \geq t_0, \end{array}$$

for every solution $\psi(t)$ of (1). Otherwise, $\phi(t)$ is unstable.

There is a stronger version of *stability* as *asymptotically stable*

Definition 4.1.2 (Asymptotically stable).
A solution $\mathbf{x} = \phi(t)$ of (1) is asymptotically stable if it is stable and every solution $\psi(t)$ which starts sufficiently close to $\phi(t)$ satisfies

$$\lim_{t\to\infty} \big(\psi_j(t) - \phi_j(t)\big) = 0, \qquad j = 1, \ldots, n.$$

Definition 4.1.3 (Equilibrium solution)**.**
A point $\mathbf{x}^0$ *satisfying*

$$f(\mathbf{x}^0) = \mathbf{0}$$

is called an equilibrium point. The constant function

$$\mathbf{x}(t) = \mathbf{x}^0$$

is the corresponding equilibrium solution.

We first consider an easier case:

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = A\mathbf{x}$$

This will be talked about in 1.1.

1.1 Stability for linear systems

Consider

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = A\mathbf{x} \tag{2}$$

We first show:

Stability of any solution of (2) is determined by stability of $\mathbf{x}(t) = \mathbf{0}$.

Easy to notice that $\mathbf{x}(t) = 0$ is obviously a solution for (2). Also, any two solutions $\mathbf{y}_1$ and $\mathbf{y}_2$: we have $\mathbf{y}_3 = \mathbf{y}_1 - \mathbf{y}_2$ is also a solution of (2).

Consider any solution of (2) as $\phi(t)$: if we want to analysis its stability, we want to study $\psi(t)$ which has:

$$|\psi_j(0) - \phi_j(0)| < \delta, \quad j = 1, \ldots, n \tag{3}$$

and we want to analysis

$$|\psi_j(t) - \phi_j(t)| \tag{4}$$

Set

$$\mathbf{z}(t) = \psi(t) - \phi(t).$$

Then $\mathbf{z}(t)$ is also a solution of (2).
Also, if $\psi(0)$ is close to $\phi(0)$, then $\mathbf{z}(0)$ is close to $\mathbf{0}$. Thus, the problem becomes the stability of the equilibrium solution $\mathbf{x}(t) = \mathbf{0}$.

Conversely, if $\mathbf{x}(t) = \mathbf{0}$ is unstable, then there exists a solution $\mathbf{h}(t)$ which starts arbitrarily close to $\mathbf{0}$ but later leaves a fixed neighborhood of $\mathbf{0}$. For any solution $\phi(t)$,

$$\psi(t) = \phi(t) + \mathbf{h}(t)$$

is another solution. It starts close to $\phi(t)$, but it does not remain close to $\phi(t)$. Therefore,

$$\boxed{\begin{gathered}\phi(t) \text{ is stable} \iff \mathbf{0} \text{ is stable,} \\ \phi(t) \text{ is asymptotically stable} \iff \mathbf{0} \text{ is asymptotically stable.}\end{gathered}} \tag{5}$$

We now reduce the n component estimates to one estimate.

Definition 4.1.4 (Infinity norm)**.**
For

$$\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{C}^n,$$

define

$$\boxed{\|\mathbf{x}\|_\infty = \max_{1\le j\le n} |x_j|.} \tag{6}$$

This norm satisfies

$$\|\mathbf{x}\|_\infty \ge 0, \qquad \|\mathbf{x}\|_\infty = 0 \iff \mathbf{x} = \mathbf{0}, \quad \|c\mathbf{x}\|_\infty = |c|\,\|\mathbf{x}\|_\infty, \quad \|\mathbf{x}+\mathbf{y}\|_\infty \le \|\mathbf{x}\|_\infty + \|\mathbf{y}\|_\infty.$$

Theorem 4.1.1 (Stability of linear systems)**.** *Consider*

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = A\mathbf{x}.$$

1. *If every eigenvalue of A has negative real part, then every solution is asymptotically stable.*
2. *If A has an eigenvalue with positive real part, then every solution is unstable.*
3. *Suppose every eigenvalue has nonpositive real part. For every eigenvalue on the imaginary axis, let its algebraic multiplicity be k. Every solution is stable if A has k linearly independent eigenvectors for each such eigenvalue. Otherwise, every solution is unstable.*

1.1.1 Proof of Theorem 4.1.1

Proof. By (5), it is enough to study $\mathbf{x}(t) = \mathbf{0}$.
Case 1: every eigenvalue has negative real part.
Let

$$-\alpha_0 = \max_{\lambda\in\sigma(A)} \operatorname{Re}\lambda, \qquad \alpha_0 > 0.$$

By the generalized-eigenvector formula (23), every entry of e^{At} is a finite linear combination of terms of the form

$$t^r e^{\lambda t}, \qquad r \ge 0.$$

Choose $0 < \alpha < \alpha_0$. Since

$$t^r e^{-(\alpha_0-\alpha)t}$$

is bounded for $t \ge 0$, there exists $K > 0$ such that

$$\left|(e^{At})_{ij}\right| \le K e^{-\alpha t}, \qquad t \ge 0.$$

Every solution has the form

$$\mathbf{x}(t) = e^{At}\mathbf{x}(0).$$

So

$$\boxed{\|\mathbf{x}(t)\|_\infty \le nKe^{-\alpha t}\|\mathbf{x}(0)\|_\infty.} \tag{7}$$

Given $\varepsilon > 0$, choose

$$\delta(\varepsilon) = \frac{\varepsilon}{nK}.$$

Then

$$\|\mathbf{x}(0)\|_\infty < \delta \quad \Longrightarrow \quad \|\mathbf{x}(t)\|_\infty < \varepsilon \qquad (t \ge 0).$$

Thus, the zero solution is stable. Moreover,

$$\lim_{t\to\infty} \|\mathbf{x}(t)\|_\infty = 0,$$

so it is asymptotically stable.
Case 2: an eigenvalue has positive real part.
First suppose

$$\lambda > 0$$

is real, with real eigenvector $\mathbf{v}$. For any $c \ne 0$,

$$\mathbf{x}(t) = ce^{\lambda t}\mathbf{v}$$

is a solution and

$$\|\mathbf{x}(t)\|_\infty = |c|e^{\lambda t}\|\mathbf{v}\|_\infty \longrightarrow \infty.$$

The number c can be arbitrarily small, so the zero solution is unstable.
Now suppose

$$\lambda = \alpha + i\beta, \qquad \alpha > 0,$$

with eigenvector

$$\mathbf{v} = \mathbf{p} + i\mathbf{q}.$$

The real part of $e^{\lambda t}\mathbf{v}$ is

$$\mathbf{x}_1(t) = e^{\alpha t}\big(\mathbf{p}\cos(\beta t) - \mathbf{q}\sin(\beta t)\big).$$

The imaginary part is also a real-valued solution. Since $\mathbf{p}$ and $\mathbf{q}$ are not both zero, at least one of these two real solutions is unbounded. Multiplying it by an arbitrarily small constant gives instability.
Case 3: every eigenvalue has nonpositive real part.
Let $\lambda_j = i\sigma_j$ be an eigenvalue on the imaginary axis. If it has as many linearly independent eigenvectors as its algebraic multiplicity, then no positive power of t occurs and

$$\left|e^{i\sigma_j t}\right| = 1.$$

All other terms decay. Thus, for some $K > 0$,

$$\left|(e^{At})_{ij}\right| \le K, \qquad t \ge 0.$$

$$\boxed{\|\mathbf{x}(t)\|_\infty \le nK\|\mathbf{x}(0)\|_\infty.} \tag{8}$$

$$\boxed{\delta(\varepsilon) = \frac{\varepsilon}{nK}}$$

proves stability.

If $\lambda = i\sigma$ does not have enough independent eigenvectors, choose $\mathbf{v}$ such that

$$(A - i\sigma I)^2\mathbf{v} = \mathbf{0}, \qquad \mathbf{w} := (A - i\sigma I)\mathbf{v} \neq \mathbf{0}.$$

Then (23) gives

$$\mathbf{x}(t) = e^{i\sigma t}\big(\mathbf{v} + t\mathbf{w}\big). \tag{9}$$

If $\sigma = 0$, we may choose a real generalized eigenvector, and the solution is unbounded. If $\sigma \neq 0$, at least one of the real and imaginary parts of (9) is an unbounded real solution. Scaling its initial value proves instability. $\square$

1.2 Stability for *almost* linear systems

Before, we considered

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = A\mathbf{x}.$$

Now consider

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = A\mathbf{x} + g(\mathbf{x}), \tag{10}$$

where

$$g(\mathbf{x}) = \begin{bmatrix} g_1(\mathbf{x}) \\ \vdots \\ g_n(\mathbf{x}) \end{bmatrix}, \qquad g(\mathbf{0}) = \mathbf{0}. \tag{11}$$

Thus, $\mathbf{x}(t) = \mathbf{0}$ is an equilibrium solution of (10).
We want $g(\mathbf{x})$ to be small compared with $\mathbf{x}$. For $\mathbf{x} \neq \mathbf{0}$, set

$$F(\mathbf{x}) = \frac{g(\mathbf{x})}{\|\mathbf{x}\|_\infty}, \qquad F(\mathbf{0}) = \mathbf{0}. \tag{12}$$

The condition is

$$F \text{ is continuous at } \mathbf{0} \quad \Longleftrightarrow \quad \lim_{\mathbf{x}\to\mathbf{0}} \frac{\|g(\mathbf{x})\|_\infty}{\|\mathbf{x}\|_\infty} = 0.$$

For example, this holds if every component of $g(\mathbf{x})$ begins with terms of order two or higher.

Theorem 4.1.2. *Suppose that* (12) *is continuous at* $\mathbf{0}$*, and that the initial-value problem for* (10) *is locally well posed.*

1. *If every eigenvalue of A has negative real part, then $\mathbf{x}(t) = \mathbf{0}$ is asymptotically stable.*
2. *If A has an eigenvalue with positive real part, then $\mathbf{x}(t) = \mathbf{0}$ is unstable.*
3. *If every eigenvalue has nonpositive real part and at least one eigenvalue has zero real part, then the linearized equation alone does not determine the stability.*

1.2.1 Proof of Theorem 4.1.2

Suppose every eigenvalue of A has negative real part. From Case 1 of Theorem 4.1.1, there exist $K, \alpha > 0$ such that

$$\|e^{At}\mathbf{v}\|_\infty \le Ke^{-\alpha t}\|\mathbf{v}\|_\infty, \qquad t \ge 0. \tag{13}$$

Using (36) with $\mathbf{f}(t) = g(\mathbf{x}(t))$, we have

$$\mathbf{x}(t) = e^{At}\mathbf{x}(0) + \int_0^t e^{A(t-s)} g(\mathbf{x}(s))\,\mathrm{d}s. \tag{14}$$

By (12), there exists $\rho > 0$ such that

$$\|g(\mathbf{x})\|_\infty \le \frac{\alpha}{2K}\|\mathbf{x}\|_\infty, \qquad \|\mathbf{x}\|_\infty < \rho. \tag{15}$$

As long as

$$\|\mathbf{x}(s)\|_\infty < \rho, \qquad 0 \le s \le t,$$

(13), (14), and (15) give

$$\|\mathbf{x}(t)\|_\infty \le Ke^{-\alpha t}\|\mathbf{x}(0)\|_\infty + \frac{\alpha}{2}\int_0^t e^{-\alpha(t-s)}\|\mathbf{x}(s)\|_\infty\,\mathrm{d}s. \tag{16}$$

Multiplying by $e^{\alpha t}$, set

$$z(t) = e^{\alpha t}\|\mathbf{x}(t)\|_\infty.$$

Then

$$z(t) \le K\|\mathbf{x}(0)\|_\infty + \frac{\alpha}{2}\int_0^t z(s)\,\mathrm{d}s. \tag{17}$$

We cannot differentiate the two sides of an inequality directly. Set

$$U(t) = \frac{\alpha}{2}\int_0^t z(s)\,\mathrm{d}s.$$

By (17),

$$\begin{aligned} U'(t) &= \frac{\alpha}{2} z(t) \\ &\leq \frac{\alpha K}{2}\|\mathbf{x}(0)\|_\infty + \frac{\alpha}{2}U(t), \end{aligned}$$

so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[e^{-\alpha t/2}\left(U(t) + K\|\mathbf{x}(0)\|_\infty\right)\right] \leq 0.$$

Therefore,

$$e^{-\alpha t/2}\left(U(t) + K\|\mathbf{x}(0)\|_\infty\right) \leq K\|\mathbf{x}(0)\|_\infty,$$

and

$$U(t) + K\|\mathbf{x}(0)\|_\infty \leq K\|\mathbf{x}(0)\|_\infty e^{\alpha t/2}.$$

Returning to (17),

$$\boxed{\|\mathbf{x}(t)\|_\infty \leq K\|\mathbf{x}(0)\|_\infty e^{-\alpha t/2}.} \tag{18}$$

We still need to know that $\|\mathbf{x}(t)\|_\infty < \rho$. Suppose $T > 0$ is the first time that

$$\|\mathbf{x}(T)\|_\infty = \rho.$$

If

$$\|\mathbf{x}(0)\|_\infty < \frac{\rho}{K},$$

then (18) gives

$$\|\mathbf{x}(T)\|_\infty \leq K\|\mathbf{x}(0)\|_\infty e^{-\alpha T/2} < \rho,$$

a contradiction. Hence, (18) holds for every $t \geq 0$.
Given $\varepsilon > 0$, choose

$$\delta(\varepsilon) = \min\left\{\frac{\rho}{K}, \frac{\varepsilon}{K}\right\}.$$

Then

$$\|\mathbf{x}(0)\|_\infty < \delta \quad\Longrightarrow\quad \|\mathbf{x}(t)\|_\infty < \varepsilon, \qquad t \geq 0,$$

and

$$\lim_{t\to\infty}\|\mathbf{x}(t)\|_\infty = 0.$$

Thus, $\mathbf{x}(t) = \mathbf{0}$ is asymptotically stable. □

1.2.2 When an eigenvalue has zero real part

The nonlinear term may now determine the stability. Consider the two systems

$$\begin{aligned} \frac{\mathrm{d}x_1}{\mathrm{d}t} &= x_2 \mp x_1(x_1^2 + x_2^2), \\ \frac{\mathrm{d}x_2}{\mathrm{d}t} &= -x_1 \mp x_2(x_1^2 + x_2^2). \end{aligned} \tag{19}$$

Both have the same linearized matrix

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \lambda_{1,2} = \pm i.$$

Set

$$R(t) = x_1^2(t) + x_2^2(t).$$

For the upper signs,

$$\begin{aligned} R'(t) &= 2x_1x_1' + 2x_2x_2' \\ &= -2(x_1^2 + x_2^2)^2 \\ &= -2R^2(t), \end{aligned}$$

so

$$\boxed{R(t) = \frac{R(0)}{1 + 2R(0)t} \longrightarrow 0.}$$

Thus, the zero solution is asymptotically stable.
For the lower signs,

$$R'(t) = 2R^2(t),$$

and

$$\boxed{R(t) = \frac{R(0)}{1 - 2R(0)t}.}$$

Every nonzero solution leaves a fixed neighborhood of the origin and becomes unbounded in finite time. Thus, the zero solution is unstable.

1.3 Stability of equilibrium solutions

Consider

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = f(\mathbf{x}), \tag{20}$$

and let $\mathbf{x}^0$ be an equilibrium point:

$$f(\mathbf{x}^0) = \mathbf{0}.$$

Set

$$\mathbf{z}(t) = \mathbf{x}(t) - \mathbf{x}^0. \tag{21}$$

Then

$$\frac{\mathrm{d}\mathbf{z}}{\mathrm{d}t} = f(\mathbf{x}^0 + \mathbf{z}),$$

and

$$\mathbf{x}(t) = \mathbf{x}^0 \text{ is stable} \quad \Longleftrightarrow \quad \mathbf{z}(t) = \mathbf{0} \text{ is stable.}$$

The same equivalence holds for asymptotic stability.
We want to write

$$f(\mathbf{x}^0 + \mathbf{z}) = A\mathbf{z} + g(\mathbf{z}).$$

Lemma 1.3 (Taylor expansion at an equilibrium)**.** *Suppose f has continuous second-order partial derivatives near $\mathbf{x}^0$. Then*

$$f(\mathbf{x}^0 + \mathbf{z}) = f(\mathbf{x}^0) + Df(\mathbf{x}^0)\mathbf{z} + g(\mathbf{z}), \tag{22}$$

where

$$\lim_{\mathbf{z}\to\mathbf{0}} \frac{\|g(\mathbf{z})\|_\infty}{\|\mathbf{z}\|_\infty} = 0.$$

Proof. For $j = 1, \dots, n$, Taylor's theorem gives

$$f_j(\mathbf{x}^0 + \mathbf{z}) = f_j(\mathbf{x}^0) + \sum_{\ell=1}^{n} \frac{\partial f_j}{\partial x_\ell}(\mathbf{x}^0) z_\ell + g_j(\mathbf{z}),$$

where

$$|g_j(\mathbf{z})| \le C\|\mathbf{z}\|_\infty^2$$

for $\mathbf{z}$ sufficiently small. Hence,

$$\|g(\mathbf{z})\|_\infty \le C\|\mathbf{z}\|_\infty^2,$$

and

$$0 \le \frac{\|g(\mathbf{z})\|_\infty}{\|\mathbf{z}\|_\infty} \le C\|\mathbf{z}\|_\infty \longrightarrow 0.$$

Collecting the component equations gives (22), with

$$\boxed{A = Df(\mathbf{x}^0) = \begin{pmatrix} \dfrac{\partial f_1}{\partial x_1}(\mathbf{x}^0) & \cdots & \dfrac{\partial f_1}{\partial x_n}(\mathbf{x}^0) \\ \vdots & \ddots & \vdots \\ \dfrac{\partial f_n}{\partial x_1}(\mathbf{x}^0) & \cdots & \dfrac{\partial f_n}{\partial x_n}(\mathbf{x}^0) \end{pmatrix}.} \tag{23}$$

□

Since $f(\mathbf{x}^0) = \mathbf{0}$, the shifted equation becomes

$$\frac{\mathrm{d}\mathbf{z}}{\mathrm{d}t} = Df(\mathbf{x}^0)\mathbf{z} + g(\mathbf{z}).$$

Therefore, Theorem 4.1.2 gives the following result.

Theorem 4.1.4 (Linearization at an equilibrium)**.** *Let f satisfy the hypotheses of Lemma 1.3, and suppose the initial-value problem is locally well posed near $\mathbf{x}^0$.*

1. *If every eigenvalue of $Df(\mathbf{x}^0)$ has negative real part, then $\mathbf{x}(t) = \mathbf{x}^0$ is asymptotically stable.*
2. *If $Df(\mathbf{x}^0)$ has an eigenvalue with positive real part, then $\mathbf{x}(t) = \mathbf{x}^0$ is unstable.*
3. *If every eigenvalue has nonpositive real part and at least one has zero real part, then this test is inconclusive.*

Thus, the procedure is

$$\boxed{\begin{aligned} &f(\mathbf{x}^0) = \mathbf{0}, \\ &\mathbf{z} = \mathbf{x} - \mathbf{x}^0, \\ &A = Df(\mathbf{x}^0), \\ &\sigma(A) \implies \begin{cases} \operatorname{Re}\lambda < 0 & \implies \text{asymptotically stable}, \\ \operatorname{Re}\lambda > 0 \text{ for some } \lambda & \implies \text{unstable}, \\ \operatorname{Re}\lambda \le 0 \text{ with some } \operatorname{Re}\lambda = 0 & \implies \text{inconclusive}. \end{cases} \end{aligned}}$$